EUCLIDEAN RAMSEY THEORY FOR LINES AND FINITE CONFIGURATIONS

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Abstract. A classical question in Euclidean Ramsey theory, posed by Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus, asks which finite configurations $X \subset \mathbb{E}^2$ must appear in blue under every red–blue coloring of the plane having no red pair of points distance one apart. For $m \geq 2$, let ℓ_m denote the line of m points with consecutive points distance one apart. Tsaturian showed that ℓ_5 is unavoidable; we improve this to ℓ_7. Additionally, we show that in every three-coloring of the plane in which two colors contain no ℓ_2, the third color must contain an ℓ_3. To conclude our discussion on lines, we prove several related results on two-colorings of higher-dimensional Euclidean spaces where both colors avoid longer lines. Our methods also extend to other point configurations. In any two-coloring of the plane in which one color contains no ℓ_2, we show that the other color must contain both a regular pentagon of side length one and a unit square together with its center. We also prove that every two-coloring of three-dimensional space contains either a monochromatic unit square or a monochromatic noncoplanar tetrahedron with four edges of length one and two of length $\sqrt{2}$. Our proofs combine Székely's averaging inequalities for distance-avoiding sets with the Fourier-analytic and linear-programming methods of de Oliveira Filho and Vallentin, together with a theorem on rotation-invariant measures from OpenAI's recent work on the Hadwiger-Nelson problem.

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1. Introduction

Euclidean Ramsey theory, initiated in a series of papers by Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus, asks which finite configurations are unavoidable in every coloring of Euclidean space [16, 17, 18]. We write $\mathbb{E}^d$ for $\mathbb{R}^d$ equipped with its Euclidean metric. For two configurations $X, Y \subset \mathbb{E}^d$, we write $\mathbb{E}^d \to (X, Y)$ if every red–blue coloring of $\mathbb{E}^d$ contains either a red X or a blue Y (up to isometry). Its negation, denoted $\mathbb{E}^d \not\to (X, Y)$, means that there exists a red–blue coloring with no red X and no blue Y. More generally, for configurations $X_1, \dots, X_r \subset \mathbb{E}^d$, write $\mathbb{E}^d \to (X_1, \dots, X_r)$ if every r-coloring of $\mathbb{E}^d$ contains a configuration isometric to X_i in some color i. Therefore $\mathbb{E}^d \not\to (X_1, \dots, X_r)$ means that there exists an r-coloring in which color i contains no configuration isometric to X_i for every i.

1.1. **Lines.** For $m \geq 2$, let $\ell_m = \{0, 1, \dots, m-1\} \subset \mathbb{R}$ be the unit-spaced m-point line. Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus proved $\mathbb{E}^2 \to (\ell_2, \ell_4)$ and also $\mathbb{E}^2 \to (\ell_2, K)$ for every three-point configuration $K \subset \mathbb{E}^2$ [17].

Subsequent work has extended these results in several directions. Tsaturian later advanced $\mathbb{E}^2 \to (\ell_2, \ell_4)$ to $\mathbb{E}^2 \to (\ell_2, \ell_5)$ [33]. In dimension 3, Iván established $\mathbb{E}^3 \to (\ell_2, K)$ for every five-point configuration $K \subset \mathbb{E}^3$, while Arman and Tsaturian showed $\mathbb{E}^3 \to (\ell_2, \ell_6)$ [25, 2]. An argument of Szlam gives, for some constant $c' > 0$, $\mathbb{E}^d \to (\ell_2, K)$ for every $K \subset \mathbb{E}^d$ with $|K| \leq 2^{c'd}$ [32, 6].

Our first result strengthens the lower bound (of Tsaturian) in the plane:

Theorem 1. $\mathbb{E}^2 \to (\ell_2, \ell_7)$.

In the opposite direction, Erdős and Graham claimed, without proof, that $\mathbb{E}^2 \not\to (\ell_2, \ell_m)$ for some $m < 10^7$ [19]. Conlon and Fox proved $\mathbb{E}^2 \not\to (\ell_2, \ell_{10^{10}})$ and, for some constant $c > 0$, $\mathbb{E}^d \not\to (\ell_2, \ell_m)$ whenever $d \geq 2$ and $m \geq 2^{cd}$ [6] by a nonconstructive argument. Recently, Currier, Mody, Xie, and Zhang improved the planar bound to $\mathbb{E}^2 \not\to (\ell_2, \ell_{6330})$. Their proof follows the general outline of Conlon and Fox, using lattice-based colorings and probabilistic arguments, and is also nonconstructive [10]. In forthcoming work, the second and third authors of this paper show that $\mathbb{E}^2 \not\to (\ell_2, \ell_{42})$ by a computer-assisted explicit construction [24]. Combining these results leaves the cases $8 \leq m \leq 41$ unresolved.

We next consider colorings with three or more colors. The Hadwiger–Nelson problem asks for the chromatic number of the graph whose vertices are the points of the plane and whose edges join pairs of points distance one apart. Such a coloring has no monochromatic ℓ_2, so lower bounds for this chromatic number imply corresponding results with repeated ℓ_2. The Moser spindle shows that this graph requires at least four colors, and hence $\mathbb{E}^2 \to (\ell_2, \ell_2, \ell_2)$ [27]. As a corollary of de Grey's result that the chromatic number of the plane is at least five [13], every four-coloring contains a monochromatic ℓ_2; that is, $\mathbb{E}^2 \to (\ell_2, \ell_2, \ell_2, \ell_2)$. OpenAI proved, using entirely new techniques, that the plane is not five-colorable, raising the lower bound in the Hadwiger–Nelson problem to six [29]. Consequently, $\mathbb{E}^2 \to (\ell_2, \ell_2, \ell_2, \ell_2, \ell_2)$. For three colors, Currier, Moore, and Yip proved that, for every $d \geq 2$, $\mathbb{E}^d \not\to (\ell_3, \ell_3, \ell_8)$, $\mathbb{E}^d \not\to (\ell_3, \ell_4, \ell_7)$, and $\mathbb{E}^d \not\to (\ell_3, \ell_5, \ell_5)$ [12]. Theorem 2 strengthens the three-color conclusion from the Moser spindle by replacing one of the configurations ℓ_2 with ℓ_3.

Theorem 2. $\mathbb{E}^2 \to (\ell_2, \ell_2, \ell_3)$.

We now return to two colors and consider lines in higher dimensions. For two-color problems involving a red ℓ_3, Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus proved $\mathbb{E}^3 \to (K, K)$ for every three-point configuration $K \subset \mathbb{E}^3$, and hence $\mathbb{E}^3 \to (\ell_3, \ell_3)$ [16]. In the plane, Currier, Moore, and Yip later proved $\mathbb{E}^2 \to (\ell_3, \ell_3)$ with computer assistance [11]. Conlon and Wu proved $\mathbb{E}^d \not\to (\ell_3, \ell_{10^{50}})$ for every $d \geq 2$ [8]. Führer and Tóth improved 10^{50} to 1177 [22], and Currier, Moore, and Yip later improved it to 20 [12]. Thus, in the plane, there remains a gap $4 \leq m \leq 19$. We were unable to improve the bounds in the plane, but in higher dimensions, we prove:

Theorem 3. $\mathbb{E}^3 \to (\ell_3, \ell_4)$.

Theorem 4. $\mathbb{E}^4 \to (\ell_3, \ell_5)$.

Theorem 5. $\mathbb{E}^7 \to (\ell_3, \ell_6)$.

Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus proved $\mathbb{E}^d \not\to (\ell_6, \ell_6)$ and $\mathbb{E}^d \not\to (\ell_4, \ell_4, \ell_4)$ for every $d \geq 2$ [16]. For two colors, interchanging the colors in $\mathbb{E}^2 \to (\ell_2, \ell_4)$ gives $\mathbb{E}^2 \to (\ell_4, \ell_2)$ [17]. Currier, Moore, and Yip proved $\mathbb{E}^d \not\to (\ell_4, \ell_{14})$ and $\mathbb{E}^d \not\to (\ell_5, \ell_8)$ for every $d \geq 2$ [12]. Thus the planar gap for red ℓ_4 is $3 \leq m \leq 13$. In dimension 4, we prove:

Theorem 6. $\mathbb{E}^4 \to (\ell_4, \ell_4)$.

Table 1 compares Theorems 1–6 with the results discussed above.

Theorem	Prior lower bound	New lower bound	Upper bound
1	$\mathbb{E}^2 \to (\ell_2, \ell_5)$	$\mathbb{E}^2 \to (\ell_2, \ell_7)$	$\mathbb{E}^2 \not\to (\ell_2, \ell_{42})$
2	$\mathbb{E}^2 \to (\ell_2, \ell_2, \ell_2)$	$\mathbb{E}^2 \to (\ell_2, \ell_2, \ell_3)$	
3		$\mathbb{E}^3 \to (\ell_3, \ell_4)$	
4	$\mathbb{E}^2 \to (\ell_3, \ell_3)$	$\mathbb{E}^4 \to (\ell_3, \ell_5)$	$\mathbb{E}^d \not\to (\ell_3, \ell_{20})$
5		$\mathbb{E}^7 \to (\ell_3, \ell_6)$	
6	$\mathbb{E}^2 \to (\ell_4, \ell_2)$	$\mathbb{E}^4 \to (\ell_4, \ell_4)$	$\mathbb{E}^d \not\to (\ell_4, \ell_{14})$

TABLE 1. Line results.

It is also worth mentioning a more general result of Conlon and Führer: if a finite configuration $X \subset \mathbb{E}^d$ is not contained in a sphere, then for some integer m, $\mathbb{E}^n \not\to (X, \ell_m)$ for every $n \geq d$ [7].

1.2. **Other configurations.** For $n \geq 3$ and $s > 0$, write $W_n(s)$ for a regular n-gon of side length s together with its center, and abbreviate $W_4(1)$ to W_4. Juhász [26] showed $\mathbb{E}^2 \to (\ell_2, K)$ for every four-point subset $K \subset \mathbb{E}^2$, which immediately gives $\mathbb{E}^2 \to (\ell_2, W_3(s))$ for all $s > 0$, and constructed a 12-point set $K \subset \mathbb{E}^2$ with $\mathbb{E}^2 \not\to (\ell_2, K)$. Csizmadia and Tóth [9] cut twelve points down to eight with a centered heptagon of side $s = \frac{9}{5}\sin(\pi/7)$, proving $\mathbb{E}^2 \not\to \left(\ell_2, W_7(\frac{9}{5}\sin(\pi/7))\right)$. Among the open cases $4 \leq n \leq 6$, we settle W_4:

Theorem 7. $\mathbb{E}^2 \to (\ell_2, W_4)$.

Removing the center, let P_n ($n \geq 3$) denote the regular n-gon of unit side length. $\mathbb{E}^2 \to (\ell_2, P_4)$ is a corollary of Juhász's theorem, and $\mathbb{E}^2 \to (\ell_2, P_6)$ is immediate: if a red point x exists, the six vertices of a unit regular hexagon centered at x must all be blue, whereas if no point is red the entire plane is blue. For the regular pentagon P_5, we have:

Theorem 8. $\mathbb{E}^2 \to (\ell_2, P_5)$.

Let e_1, e_2, e_3 form an orthonormal basis of $\mathbb{R}^3$, and define $\mathcal{S} = \{0, e_1, e_2, e_1 + e_2\}$ and $\mathcal{T} = \left\{0, e_1, e_2, \frac{1}{2}e_1 + \frac{\sqrt{3}}{2}e_3\right\}$. These are a unit square and a noncoplanar tetrahedron. Each has four pairwise distances equal to 1 and two equal to $\sqrt{2}$. A folklore construction gives $\mathbb{E}^2 \nrightarrow (\mathcal{S}, \mathcal{S})$ [23]. Burr's argument gives $\mathbb{E}^6 \to (\mathcal{S}, \mathcal{S})$ [16], and Cantwell lowered the dimension to 4 [5]. We later show that every four-point set in $\mathbb{E}^3$ with distance multiset $\{1, 1, 1, 1, \sqrt{2}, \sqrt{2}\}$ is isometric to either $\mathcal{S}$ or $\mathcal{T}$. We prove:

Theorem 9. *Every two-coloring of* $\mathbb{E}^3$ *contains a monochromatic* $\mathcal{S}$ *or* $\mathcal{T}$.

1.3. **Methods.** We briefly sketch a proof of $\mathbb{E}^2 \to (\ell_2, \ell_7)$ (Theorem 1) to illustrate our proof method. We first establish an inequality for colorings of a finite configuration, then average it and show that its expected value is negative. Let P be the vertex set of an 18-point triangular lattice strip (see Figure 1 in Section 3). Let $r_v \in \{0, 1\}$ indicate whether vertex $v \in P$ is red. We construct a quadratic polynomial $\mathcal{L}((r_v)_{v \in P})$ in these 18 variables that is nonnegative for every coloring of P containing no red ℓ_2 and no blue ℓ_7. Its coefficients are chosen by linear programming. To extend this local inequality to colorings of the entire plane, we average it over all translates of the configuration P.

To average this inequality without assuming measurability, start with a coloring of $\mathbb{E}^2$ with no red ℓ_2 and no blue ℓ_7. Restrict it to $\mathbb{F}^2$, where $\mathbb{F} = \mathbb{R} \cap \overline{\mathbb{Q}}$ is the field of real algebraic numbers, and let $\mathcal{X}$ be the space of such colorings of $\mathbb{F}^2$. We show that $\mathcal{X}$ is compact and metrizable, and the group $G = \mathbb{F}^2 \rtimes \mathrm{SO}(2, \mathbb{F})$ acts on $\mathcal{X}$. Furthermore, this group is solvable and therefore amenable, so we can average our restricted coloring over a Følner sequence in G. Since $\mathcal{X}$ is compact, Prokhorov's theorem implies that a subsequence of these averages converges weakly to a probability measure μ, which is invariant under G. Since (by construction) $\mathcal{L}$ depends only on a finite number of points, it is continuous on $\mathcal{X}$, and we can show that $\mathbb{E}_\mu[\mathcal{L}] \geq 0$.

Now we aim to obtain an upper bound for this expected value. By translation invariance, $\mathbb{E}_\mu[r_v] = \mathbb{E}_\mu[r_0]$ and $\mathbb{E}_\mu[r_u r_v] = C(v - u)$, where $C(z) = \mathbb{E}_\mu[r_0 r_z]$. The correlation function is positive definite, so Bochner's theorem gives

$$C(z) = \int_{\widehat{\mathbb{F}^2}} \chi(z)\, d\lambda(\chi),$$

where λ is a positive $\mathrm{SO}(2, \mathbb{F})$-invariant measure. So instead of studying the original coloring, we can bound $\mathbb{E}_\mu[\mathcal{L}]$ through its Fourier representation.

However, there is still some difficulty, as not every character of $\mathbb{F}^2$ extends continuously to $\mathbb{R}^2$. Recent work by [29] shows that the part of λ supported on these discontinuous characters is a multiple of Haar measure, whose Fourier coefficients vanish at every

nonzero displacement. Therefore, we can simply focus on the continuous characters $z \mapsto e^{i\xi\cdot z}$, with $\xi \in \mathbb{R}^2$, whose rotational averages give Bessel integrals and allow us to bound the pair correlations appearing in $\mathbb{E}_\mu[\mathcal{L}]$.

For our choice of coefficients, the resulting bound gives $\mathbb{E}_\mu[\mathcal{L}] \le -1$ using Székely's averaging inequalities and the Fourier methods of de Oliveira Filho and Vallentin [31, 14]. This contradicts $\mathbb{E}_\mu[\mathcal{L}] \ge 0$. Our main contribution is to construct local inequalities for the configurations considered here and apply them to arbitrary colorings (rather than measurable colorings). The other arguments use different finite configurations but follow a similar approach. In particular, the W_4 case requires an additional bound $\mathbb{E}_\mu[r_0] \le 1/4$, and in dimensions $d \ge 3$ we use spherical averages.

2. Fourier analysis of invariant measures

2.1. **Invariant probability measures on colorings.** Let $\mathbb{F} = \mathbb{R} \cap \overline{\mathbb{Q}}$. Since all finite configurations P that appear in our local inequalities lie in $\mathbb{F}^d$, we can restrict any coloring of $\mathbb{E}^d$ avoiding the forbidden configurations to one on $\mathbb{F}^d$. For k colors with forbidden configurations assigned to each $i \in \{1,\dots,k\}$, let $\mathcal{X} \subseteq \{1,\dots,k\}^{\mathbb{F}^d}$ be the set of valid colorings $\omega : \mathbb{F}^d \to \{1,\dots,k\}$. In the product topology on $\{1,\dots,k\}^{\mathbb{F}^d}$ (which is generated by cylinder sets fixing ω on finite subsets of $\mathbb{F}^d$), $\{1,\dots,k\}^{\mathbb{F}^d}$ is compact by Tychonoff's theorem and metrizable via

$$d(\omega,\eta) = \sum_{n=1}^{\infty} 2^{-n} \mathbf{1}_{\{\omega(p_n)\neq\eta(p_n)\}} \qquad (\mathbb{F}^d = \{p_1, p_2, \dots\}).$$

Note that the set $\mathcal{X}$ is closed. In fact, if $\omega \notin \mathcal{X}$, one of the forbidden configurations appears on a finite set $S \subset \mathbb{F}^d$. Every coloring agreeing with ω on S has the same issue, so the complement of $\mathcal{X}$ is open, and $\mathcal{X}$ is both compact and metrizable.

Now let's use rotations. Define $m = \lfloor d/2 \rfloor$ and $K = \{a + ib : a, b \in \mathbb{F},\ a^2 + b^2 = 1\}$, so an element $a + ib \in K$ rotates $\mathbb{F}^2$ by $(x, y) \mapsto (ax - by, bx + ay)$. The jth factor of K^m acts on $\operatorname{span}(e_{2j+1}, e_{2j+2})$ for $0 \le j < m$ and leaves the other coordinates fixed. For odd dimensions, it fixes the last coordinate as well. Together with translations, these rotations give $G_d = \mathbb{F}^d \rtimes K^m$, $(v, U)x = v + Ux$, and $(g \cdot \omega)(x) = \omega(g^{-1}x)$.

Since each $U \in K^m$ is orthogonal, $|(v + Ux) - (v + Uy)| = |x - y|$ for all $x, y \in \mathbb{F}^d$, so G_d preserves $\mathcal{X}$. Since each $g \in G_d$ permutes the coordinates of $\{1,\dots,k\}^{\mathbb{F}^d}$, the action is continuous. Now, we give G_d the discrete topology. Since G_d is an extension of the abelian group $\mathbb{F}^d$ by the abelian group K^m, it is amenable [3, Theorem G.2.1 and Proposition G.2.2(ii)]. For a countable discrete group, it is well known that amenability is equivalent to the Følner criterion: for every finite $Q \subset G_d$ and every $\varepsilon > 0$, there is a nonempty finite $F \subset G_d$ with $|gF \triangle F| < \varepsilon|F|$ for all $g \in Q$ [3, Theorem G.5.1]. Enumerating $G_d = \{g_1, g_2, \dots\}$ and applying this criterion to $Q = \{g_1, \dots, g_n\}$ with $\varepsilon = 1/n$ gives a Følner sequence of finite sets $F_n \subset G_d$ satisfying $|gF_n \triangle F_n|/|F_n| \to 0$ as $n \to \infty$ for each $g \in G_d$.

Lemma 10. *If $\mathcal{X}$ is nonempty, it supports a G_d-invariant Borel probability measure μ.*

Proof. Fix $\omega_0 \in \mathcal{X}$ and let $\nu_n = |F_n|^{-1} \sum_{g \in F_n} \delta_{g \cdot \omega_0}$ be the law of $g \cdot \omega_0$ for g drawn uniformly from F_n, where δ_ω denotes the point mass at ω. Thus $\nu_n(A)$ is the fraction of transformations in F_n whose resulting coloring belongs to the Borel set $A \subseteq \mathcal{X}$.

Probability measures on the compact metrizable space $\mathcal{X}$ are weakly sequentially compact by Prokhorov's theorem [4, Theorem 5.1]. After passing to a subsequence, ν_n therefore converges weakly to a probability measure μ. To prove invariance, fix $g \in G_d$ and a continuous function f on $\mathcal{X}$. The two averages below have identical terms on $gF_n \cap F_n$, so

$$\left| \int f(g \cdot \omega) \, d\nu_n(\omega) - \int f(\omega) \, d\nu_n(\omega) \right| = \frac{1}{|F_n|} \left| \sum_{a \in gF_n} f(a \cdot \omega_0) - \sum_{a \in F_n} f(a \cdot \omega_0) \right| \leq \|f\|_\infty \frac{|gF_n \mathbin{\triangle} F_n|}{|F_n|}.$$

As $n \to \infty$, the right-hand side vanishes by the Følner property. Since both f and $\omega \mapsto f(g \cdot \omega)$ belong to $C(\mathcal{X})$, weak convergence gives $\int f(g \cdot \omega) \, d\mu = \int f(\omega) \, d\mu$. Because $C(\mathcal{X})$ separates Borel probability measures on $\mathcal{X}$, μ is G_d-invariant. □

Fix a G_d-invariant probability measure μ on $\mathcal{X}$ and write $\mathbb{E}_\mu[f] = \int_{\mathcal{X}} f \, d\mu$. For $g \in G_d$, define $T_g f(\omega) = f(g^{-1} \cdot \omega)$ on $L^2(\mu)$, and write T_z when $g = (z, I)$ is translation by $z \in \mathbb{F}^d$. Given $\phi : \{1, \ldots, k\} \to \mathbb{R}$, set $h(\omega) = \phi(\omega(0))$, so $T_z h(\omega) = \phi(\omega(z))$, and define the pair correlation $C_h(z) = \mathbb{E}_\mu[h T_z h]$ for $z \in \mathbb{F}^d$ (when h is the indicator of color i at 0, $C_h(z)$ is the probability that both 0 and z have color i). Since μ is translation-invariant, $\mathbb{E}_\mu[(T_x h)(T_y h)] = C_h(y - x)$ for all $x, y \in \mathbb{F}^d$.

Recall that a function $F : \mathbb{F}^d \to \mathbb{C}$ is positive definite if $\sum_{i,j} \overline{c_i} c_j F(z_j - z_i) \geq 0$ for all $z_1, \ldots, z_n \in \mathbb{F}^d$ and $c_1, \ldots, c_n \in \mathbb{C}$. For C_h, we have

$$\sum_{i,j} \overline{c_i} c_j C_h(z_j - z_i) = \sum_{i,j} \overline{c_i} c_j \mathbb{E}_\mu[(T_{z_i} h)(T_{z_j} h)] = \mathbb{E}_\mu \left[\left| \sum_i c_i T_{z_i} h \right|^2 \right] \geq 0.$$

Thus C_h is positive definite, and we can apply Bochner's theorem. Also, $C_h(0) = \mathbb{E}_\mu[h^2]$, and invariance under K^m gives $C_h(Uz) = C_h(z)$ for every $U \in K^m$.

2.2. **Fourier representation in the plane.** For a countable discrete abelian group A, its Pontryagin dual $\widehat{A} = \mathrm{Hom}(A, \mathbb{T})$ (where $\mathbb{T} = \{w \in \mathbb{C} : |w| = 1\}$) is a compact metrizable abelian group under the topology of pointwise convergence. Specialized to a countable discrete group A, Bochner's theorem for locally compact abelian groups [21, Theorem 4.19] takes the following form:

Theorem 11 (Bochner)**.** *Every positive-definite function $F : A \to \mathbb{C}$ on a countable discrete abelian group A is the Fourier transform*

$$F(z) = \int_{\widehat{A}} \chi(z) \, d\lambda_F(\chi) \qquad (z \in A)$$

of a unique positive Borel measure λ_F on $\widehat{A}$, with $\lambda_F(\widehat{A}) = F(0)$.

Take $A = \mathbb{F}^2$ and set $D = \widehat{\mathbb{F}^2}$. Each $\xi \in \mathbb{R}^2$ defines a character $j(\xi) \in D$ via $j(\xi)(z) = e^{i\xi\cdot z}$, and $j : \mathbb{R}^2 \to D$ is an injective map onto the subgroup $C = j(\mathbb{R}^2)$ of characters extending continuously to $\mathbb{R}^2$. Because j maps closed balls in $\mathbb{R}^2$ homeomorphically onto compact subsets of D, the image C is Borel and $j^{-1} : C \to \mathbb{R}^2$ is Borel measurable. Under the rotation action $(u\chi)(z) = \chi(uz)$ of K on D, normalized Haar measure m_D on D satisfies $\int_D \chi(z)\, dm_D(\chi) = 0$ for all $z \in \mathbb{F}^2 \setminus \{0\}$ (as evaluation at $z \neq 0$ is a nontrivial character of D). By [29, Theorem 2.3], m_D is the unique K-invariant probability measure supported outside C:

Theorem 12. *Every K-invariant Borel probability measure on D assigning measure zero to C equals m_D.*

For $\xi \in \mathbb{R}^2$, averaging $e^{i\xi\cdot z}$ over rotations gives the Bessel function $J_0(|\xi||z|)$, where

$$J_0(t) = \frac{1}{2\pi}\int_0^{2\pi} e^{it\cos\theta}\, d\theta$$

[28, Eq. 10.9.1]. Note that if $z, w \in \mathbb{F} + i\mathbb{F}$ satisfy $|w| = |z| > 0$, then $w/z \in K$, so K acts transitively on each circle of radius $r > 0$ in $\mathbb{F}^2$. Theorems 11 and 12 give:

Lemma 13. *Every real-valued, K-invariant, positive-definite function $F : \mathbb{F}^2 \to \mathbb{R}$ admits a representation*

$$F(z) = \int_{[0,\infty)} J_0(t|z|)\, d\nu(t) + a\mathbf{1}_{\{0\}}(z) \qquad (z \in \mathbb{F}^2), \tag{2.1}$$

where ν is a finite nonnegative Borel measure on $[0,\infty)$, $a \geq 0$, and $\nu([0,\infty)) + a = F(0)$.

Proof. Let λ be the Bochner measure of F on D; by uniqueness in Theorem 11, λ and its restrictions $\lambda|_C$ and $\lambda|_{D\setminus C}$ are K-invariant. Put $a = \lambda(D \setminus C)$. If $a > 0$, then $a^{-1}\lambda|_{D\setminus C} = m_D$ by Theorem 12, so $\lambda|_{D\setminus C} = am_D$ for all $a \geq 0$, and integrating $\chi(z)$ against $\lambda|_{D\setminus C}$ gives $a\mathbf{1}_{\{0\}}(z)$.

On C, the pushforward $\rho = (j^{-1})_*(\lambda|_C)$ is a finite K-invariant Borel measure on $\mathbb{R}^2$. Since K is dense in $\mathrm{SO}(2)$, dominated convergence implies that ρ is $\mathrm{SO}(2)$-invariant. If σ is normalized Haar measure on $\mathrm{SO}(2)$, Fubini's theorem gives

$$\int_{\mathbb{R}^2} e^{i\xi\cdot z}\, d\rho(\xi) = \int_{\mathrm{SO}(2)}\int_{\mathbb{R}^2} e^{i(U\xi)\cdot z}\, d\rho(\xi)\, d\sigma(U) = \int_{\mathbb{R}^2} J_0(|\xi||z|)\, d\rho(\xi),$$

and the pushforward ν of ρ under $\xi \mapsto |\xi|$ satisfies (2.1) with $\nu([0,\infty)) = \lambda(C) = F(0) - a$. □

Placing an atom of mass a at $\infty \in [0,\infty]$ and setting $J_0(\infty \cdot |z|) = \mathbf{1}_{\{0\}}(z)$ folds $a\mathbf{1}_{\{0\}}(z)$ into (2.1): for any real function h of the color at 0 under a G_2-invariant probability measure μ, applying Lemma 13 to C_h gives a nonnegative measure $\bar{\nu}_h$ on $[0,\infty]$ with

$$C_h(z) = \int_{[0,\infty]} J_0(t|z|)\, d\bar{\nu}_h(t) \quad (z \in \mathbb{F}^2) \tag{2.2}$$

and $\bar{\nu}_h([0,\infty]) = \mathbb{E}_\mu[h^2]$.

Corollary 14. *Suppose μ is a G_2-invariant probability measure, and let r be the indicator of a single color at the origin with mean $p = \mathbb{E}_\mu[r]$. Then we can find a nonnegative Borel measure $\bar{\eta}$ on $[0,\infty]$ of total mass $\bar{\eta}([0,\infty]) = p(1-p)$ such that*

$$C_r(z) = p^2 + \int_{[0,\infty]} J_0(t|z|)\, d\bar{\eta}(t) \qquad (z \in \mathbb{F}^2). \tag{2.3}$$

Proof. We center the indicator by setting $h = r - p$. Because translation invariance gives $\mathbb{E}_\mu[T_z r] = \mathbb{E}_\mu[r] = p$, expanding the correlation directly gives $C_h(z) = \mathbb{E}_\mu[(r - p)T_z(r-p)] = C_r(z) - p^2$, while at the origin we simply have $C_h(0) = \mathbb{E}_\mu[h^2] = p(1-p)$. We then apply (2.2) to h and set $\bar{\eta} = \bar{\nu}_h$ to obtain (2.3). □

2.3. **Spherical kernels in higher dimensions.** Let σ_{d-1} be normalized surface measure on $\mathbb{S}^{d-1}$ (where $d \geq 2$) and define $\Omega_d(t) = \int_{\mathbb{S}^{d-1}} e^{-it\omega_1}\, d\sigma_{d-1}(\omega)$ for $t \in \mathbb{R}$. By [14, p. 1421], $\int_{\mathbb{S}^{d-1}} e^{ir\xi\cdot\omega}\, d\sigma_{d-1}(\omega) = \Omega_d(r|\xi|)$ for $\xi \in \mathbb{R}^d$ and $r \geq 0$. Clearly Ω_d is real and even with $\Omega_d(0) = 1$; on $[0,\infty]$ we put $\Omega_d(r\infty) = \mathbf{1}_{\{0\}}(r)$ $(r \geq 0)$. In terms of Euler's gamma function Γ, the Bessel function $J_\alpha(t)$ $(\alpha, t \geq 0)$ has the series expansion

$$J_\alpha(t) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n+\alpha+1)} \left(\frac{t}{2}\right)^{2n+\alpha}.$$

Lemma 15. *For $d \geq 2$ and $t > 0$,*

$$\Omega_d(t) = \Gamma\left(\frac{d}{2}\right)\left(\frac{2}{t}\right)^{d/2-1} J_{d/2-1}(t) = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{2^n n!\, d(d+2)\cdots(d+2n-2)},$$

where the empty product is understood to be 1. Moreover, for all $t \in \mathbb{R}$, we have the bound

$$|\Omega_d'(t)| \leq \frac{2\Gamma(d/2)}{(d-1)\sqrt{\pi}\,\Gamma((d-1)/2)}.$$

Proof. We first recall the standard Bessel formula from [14, Eq. (1)]. If we expand $J_{d/2-1}(t)$ as a power series and apply the identity $\Gamma(n + d/2) = 2^{-n} d(d+2)\cdots(d+2n-2)\Gamma(d/2)$, we can rewrite the nth term of $\Omega_d(t)$ as

$$\Gamma\left(\frac{d}{2}\right)\left(\frac{2}{t}\right)^{d/2-1} \frac{(-1)^n t^{2n+d/2-1}}{2^{2n+d/2-1} n!\,\Gamma(n+d/2)} = \frac{(-1)^n t^{2n}}{2^n n!\,\prod_{j=0}^{n-1}(d+2j)}.$$

To bound $\Omega_d'(t)$, note that for $\varepsilon \neq 0$ the difference quotients satisfy

$$\left|\frac{e^{-i(t+\varepsilon)\omega_1} - e^{-it\omega_1}}{\varepsilon}\right| \leq |\omega_1| \leq 1$$

and converge pointwise to $-i\omega_1 e^{-it\omega_1}$ as $\varepsilon \to 0$; dominated convergence therefore gives $\Omega_d'(t) = -i \int_{\mathbb{S}^{d-1}} \omega_1 e^{-it\omega_1}\, d\sigma_{d-1}(\omega)$ and hence $|\Omega_d'(t)| \leq \int_{\mathbb{S}^{d-1}} |\omega_1|\, d\sigma_{d-1}(\omega)$.

Diaconis and Freedman [15, Remark 2.10] give the density of the first k coordinates of a uniformly distributed point on the sphere of radius $\sqrt{d}$. Taking $k = 1$ and scaling the radius to one, the first coordinate of a uniformly distributed point of $\mathbb{S}^{d-1}$ has density

$$\frac{\Gamma(d/2)}{\sqrt{\pi}\,\Gamma((d-1)/2)}(1-u^2)^{(d-3)/2}$$

on $[-1,1]$. Consequently,

$$\int_{\mathbb{S}^{d-1}} |\omega_1| \, \mathrm{d}\sigma_{d-1}(\omega) = \frac{2\Gamma(d/2)}{\sqrt{\pi}\,\Gamma((d-1)/2)} \int_0^1 u(1-u^2)^{(d-3)/2} \, \mathrm{d}u$$
$$= \frac{2\Gamma(d/2)}{(d-1)\sqrt{\pi}\,\Gamma((d-1)/2)}.$$

□

For $d \in \{2,3,4,7\}$, Lemma 15 together with the half-integer Bessel identities for $J_{1/2}$ and $J_{5/2}$ [28, Eq. (10.47.3) and Sec. 10.49(i)] specializes at $t > 0$ (and by continuity at $t = 0$, where $\Omega_d(0) = 1$) to $\Omega_2(t) = J_0(t)$, $\Omega_3(t) = \frac{\sin t}{t}$, $\Omega_4(t) = \frac{2J_1(t)}{t}$, and $\Omega_7(t) = 15\frac{(3-t^2)\sin t - 3t\cos t}{t^5}$.

2.4. **Fourier representation in higher dimensions.** To apply Theorem 12 one coordinate plane at a time, we need to introduce a lemma on the product of measures.

Lemma 16. *Let Y be a separable metric space and τ a finite positive Borel measure on $Y \times D$ that is invariant under K acting on D. If $\tau(Y \times C) = 0$, then $\tau = \rho \otimes m_D$, where ρ is the Y-marginal of τ.*

Proof. For each Borel set $B \subseteq Y$, the slice measure $\tau_B(A) = \tau(B \times A)$ on D is K-invariant with $\tau_B(C) = 0$ and $\tau_B(D) = \rho(B)$. When $\rho(B) > 0$, Theorem 12 gives $\rho(B)^{-1}\tau_B = m_D$, and when $\rho(B) = 0$ we have $\tau_B = 0$. Therefore we get $\tau(B \times A) = \rho(B)m_D(A)$ for all Borel rectangles $B \times A$, and thus $\tau = \rho \otimes m_D$. □

For $d \geq 2$, let $m = \lfloor d/2 \rfloor$. Given $z \in \mathbb{F}^d$, let us write $z^{(i)} = (z_{2i-1}, z_{2i})$ for $1 \leq i \leq m$, and set $\mathcal{Z}_d = \{z \in \mathbb{F}^d : z^{(i)} \neq 0 \text{ for } 1 \leq i \leq m\}$.

Lemma 17. *Suppose h is a real function of the color at 0 under a G_d-invariant probability measure μ. Then there exists a positive Borel measure λ_h of mass at most $\mathbb{E}_\mu[h^2]$, on $\mathbb{R}^d$ if $d = 2m$ and on $\mathbb{R}^{2m} \times \widehat{\mathbb{F}}$ if $d = 2m+1$, such that for every $z \in \mathcal{Z}_d$,*

$$C_h(z) = \int_{\mathbb{R}^d} e^{i\xi\cdot z} \, d\lambda_h(\xi) \qquad \text{if } d = 2m, \tag{2.4}$$

$$C_h(v,s) = \int_{\mathbb{R}^{2m}\times\widehat{\mathbb{F}}} e^{i\xi\cdot v}\chi(s) \, d\lambda_h(\xi,\chi) \qquad \text{if } d = 2m+1, \tag{2.5}$$

where $z = (v,s)$ in the odd-dimensional case.

Proof. Let λ be the Bochner measure of C_h on D^m (if $d = 2m$) or $D^m \times \widehat{\mathbb{F}}$ (if $d = 2m+1$). Then λ has mass $\mathbb{E}_\mu[h^2]$ and is K-invariant in each of the m factors of D. For $1 \leq i \leq m$, let E_i be the set of tuples whose first D-coordinate outside C is the ith one. The sets $E_1, \ldots, E_m$ are disjoint, cover the complement of C^m (or $C^m \times \widehat{\mathbb{F}}$), and are K-invariant in the ith factor, so $\lambda|_{E_i} = \rho_i \otimes m_D$ by Lemma 16. Since $z^{(i)} \neq 0$, we have $\int_D \chi(z^{(i)}) \, dm_D(\chi) = 0$, so $\int_{E_i} \chi(z) \, d\lambda = 0$. Let λ_h denote the pushforward of $\lambda|_{C^m}$ (or $\lambda|_{C^m\times\widehat{\mathbb{F}}}$ when d is odd) under j^{-1} in each C coordinate. It is immediate that λ_h has total mass at most $\mathbb{E}_\mu[h^2]$, and by construction it satisfies (2.4)–(2.5). □

When $d \geq 3$ is odd, it is convenient to write $z = (x, y, w, s) \in \mathbb{F} \times \mathbb{F} \times \mathbb{F}^{d-3} \times \mathbb{F}$ (where w is simply empty if $d = 3$). Notice that whenever $y \neq 0$, rotating the (x, s)-plane by any $U \in K$ leaves both y and w unchanged. Therefore, the (x, y)-projection of Uz is still nonzero, and this allows us to apply (2.5) directly at Uz. With this in mind as motivation, we define $\mathcal{Z}_d^* = \{z \in \mathcal{Z}_d : z_2 \neq 0, (z_1, z_d) \neq (0, 0)\}$, on which K acts naturally by rotations in the coordinate plane spanned by e_1 and e_d.

Lemma 18. *Let $d \geq 3$ be odd, and let $h_1, \ldots, h_q$ be real functions of the color at 0 under a G_d-invariant probability measure μ. There are positive Borel measures Λ_j on $\mathbb{R}^d$, with $\Lambda_j(\mathbb{R}^d) \leq \mathbb{E}_\mu[h_j^2]$, and a Følner sequence $\Phi_n \subset K$ such that*

$$\lim_{n\to\infty} \frac{1}{|\Phi_n|} \sum_{U \in \Phi_n} C_{h_j}(Uz) = \int_{\mathbb{R}^d} e^{i\xi \cdot z} \, d\Lambda_j(\xi) \tag{2.6}$$

for every $z \in \mathcal{Z}_d^*$ and $1 \leq j \leq q$.

Proof. Let $\widetilde{\lambda}_j$ be the pushforward of λ_{h_j} onto $\mathbb{R}^{d-2} \times D$ under $(\eta_x, \eta_y, \eta_w, \chi) \longmapsto ((\eta_y, \eta_w), \psi)$, $\psi(a, b) = e^{i\eta_x a}\chi(b)$. For $z = (x, y, w, s)$ in the lemma, define the bounded continuous function

$$\Psi_z(\eta_y, \eta_w, \psi) = e^{i(\eta_y y + \eta_w \cdot w)}\psi(x, s).$$

Set $A_U(\eta, \psi) = (\eta, \psi \circ U)$ for $U \in K$. If $\Phi_n \subset K$ is a Følner sequence, we can put

$$\tau_{j,n} = \frac{1}{|\Phi_n|} \sum_{U \in \Phi_n} (A_U)_* \widetilde{\lambda}_j.$$

Then use equation (2.5) at Uz, to obtain

$$\int \Psi_z \, d\tau_{j,n} = \frac{1}{|\Phi_n|} \sum_{U \in \Phi_n} C_{h_j}(Uz).$$

Since D is compact and the $\mathbb{R}^{d-2}$-marginal is independent of n, the measures $\tau_{j,n}$ are tight and have constant mass $M_j = \widetilde{\lambda}_j(\mathbb{R}^{d-2} \times D) \leq \mathbb{E}_\mu[h_j^2]$. Now we can use Prokhorov's theorem, which tells us that a common subsequence converges weakly to measures τ_j, one for each j.

For $V \in K$ and $f \in C_b(\mathbb{R}^{d-2} \times D)$, we can exploit the Følner property to get

$$\left| \int f \circ A_V \, d\tau_{j,n} - \int f \, d\tau_{j,n} \right| \leq M_j \|f\|_\infty \frac{|V\Phi_n \triangle \Phi_n|}{|\Phi_n|} \longrightarrow 0,$$

so τ_j is K-invariant in the D coordinate. Now, Lemma 16 shows that its restriction to $\mathbb{R}^{d-2} \times (D \setminus C)$ is $\rho_j \otimes m_D$, and importantly, this part has zero integral against Ψ_z because $(x, s) \neq (0, 0)$.

A character in C has the form $\psi(a, b) = e^{i(\alpha a + \beta b)}$. Push the remaining part of τ_j forward by $(\eta_y, \eta_w, \psi) \mapsto (\alpha, \eta_y, \eta_w, \beta)$, and call the resulting measure Λ_j. Note that its mass is at most M_j. The weak convergence above gives

$$\lim_{n\to\infty} \int \Psi_z \, d\tau_{j,n} = \int \Psi_z \, d\tau_j = \int_{\mathbb{R}^d} e^{i\xi \cdot z} \, d\Lambda_j(\xi),$$

as required. □

Let $\mathrm{SO}(d,\mathbb{F})$ be the group of $d\times d$ orthogonal matrices of determinant 1 with entries in $\mathbb{F}$. For $z_1,\dots,z_N\in\mathbb{F}^d\setminus\{0\}$, real functions $h_1,\dots,h_q$ of the color at 0, and real coefficients $c, a_j, b_{j\ell}$, set $H_j(t)=a_j+\sum_{\ell=1}^N b_{j\ell}\Omega_d(|z_\ell|t)$ $(t\in[0,\infty))$ and $H_j(\infty)=a_j$.

Theorem 19. *Let $d\ge 2$ and μ be a G_d-invariant probability measure. If*

$$c+\sum_{j=1}^q a_j\mathbb{E}_\mu[h_j^2]+\sum_{j=1}^q\sum_{\ell=1}^N b_{j\ell}C_{h_j}(Rz_\ell)\ge 0\qquad \Big(R\in\mathrm{SO}(d,\mathbb{F})\Big), \tag{2.7}$$

then $c+\sum_{j=1}^q\int_{[0,\infty]}H_j(t)\,d\nu_j(t)\ge 0$ for some nonnegative Borel measures ν_j on $[0,\infty]$ with $\nu_j([0,\infty])=\mathbb{E}_\mu[h_j^2]$.

Proof. For even d, let $\rho_j=\lambda_{h_j}$ from Lemma 17. For odd d, we average (2.7) over UR, where $U\in\Phi_n$, and then take $\rho_j=\Lambda_j$ using Lemma 18. Either way, we have shown the existence of positive Borel measures ρ_j on $\mathbb{R}^d$ of total mass $M_j:=\rho_j(\mathbb{R}^d)\le\mathbb{E}_\mu[h_j^2]$ such that

$$Q(R):=c+\sum_{j=1}^q a_j\mathbb{E}_\mu[h_j^2]+\sum_{j=1}^q\sum_{\ell=1}^N b_{j\ell}\int_{\mathbb{R}^d}\cos(\xi\cdot Rz_\ell)\,d\rho_j(\xi)\ge 0 \tag{2.8}$$

whenever $R\in\mathrm{SO}(d,\mathbb{F})$ is chosen so that $Rz_1,\dots,Rz_N$ satisfy the necessary conditions of the lemma.

Since K is dense in $\mathrm{SO}(2)$ and coordinate-plane rotations generate $\mathrm{SO}(d)$, $\mathrm{SO}(d,\mathbb{F})$ is dense in $\mathrm{SO}(d)$; the projection conditions fail only on a finite union of proper closed subvarieties of $\mathrm{SO}(d)$, so $Q\ge 0$ on a dense subset of $\mathrm{SO}(d)$. Continuity of Q on $\mathrm{SO}(d)$ (by dominated convergence) then gives $Q(R)\ge 0$ on all of $\mathrm{SO}(d)$. Averaging $Q(R)$ over $R\in\mathrm{SO}(d)$ against normalized Haar measure σ via Fubini's theorem gives

$$\int_{\mathrm{SO}(d)}\int_{\mathbb{R}^d}\cos(\xi\cdot Rz_\ell)\,d\rho_j(\xi)\,d\sigma(R)=\int_{\mathbb{R}^d}\Omega_d(|\xi||z_\ell|)\,d\rho_j(\xi),$$

and setting $\nu_j=\kappa_j+(\mathbb{E}_\mu[h_j^2]-M_j)\delta_\infty$, where κ_j is the pushforward of ρ_j under $\xi\mapsto|\xi|$, gives $\nu_j([0,\infty])=\mathbb{E}_\mu[h_j^2]$ and $c+\sum_{j=1}^q\int_{[0,\infty]}H_j\,d\nu_j=\int_{\mathrm{SO}(d)}Q\,d\sigma\ge 0$. □

3. Two-colorings of the Plane

Let $P\subset\mathbb{F}^2$ be the eighteen-point triangular-lattice strip in Figure 1. Let $e_1=(1,0)$ and $e_2=(1/2,\sqrt{3}/2)$. For $i\in\{0,1\}$ and $j\in\{0,1,\dots,8\}$, put $v_{ij}=(i-1/2)e_1+(j-4)e_2$, and set $P=\{v_{ij}:i\in\{0,1\},\ j\in\{0,1,\dots,8\}\}$. We use the label ij for v_{ij}. The formula gives $|v_{ij}|\le\sqrt{73}/2$ for every vertex.

For $i\in\{0,1\}$ and $s\in\{0,1,2\}$, put $P_{i,s}=\{v_{ij}:s\le j\le s+6\}$. These are the six configurations isometric to ℓ_7 contained in P. The label difference $(\Delta i,\Delta j)$ between consecutive points in a progression in the triangular lattice satisfies $\Delta i^2+\Delta i\Delta j+\Delta j^2=1$, whose six possibilities are $(\pm1,0),(0,\pm1),(1,-1),(-1,1)$. Since $i\in\{0,1\}$, a progression of seven points must have $\Delta i=0$ and $|\Delta j|=1$, so it is one of the sets $P_{i,s}$.

For a coloring of P, let r_p be the indicator that p is red, let c_p and $a_{pq}=a_{qp}$ be the integer coefficients listed in Appendix B, and define

$$\mathcal{L}=-1+\sum_{p\in P}c_pr_p+\sum_{\{p,q\}\subset P}a_{pq}r_pr_q. \tag{3.1}$$

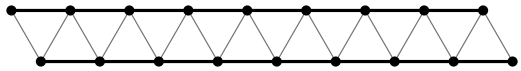


FIGURE 1. The eighteen-point triangular-lattice strip P. All drawn edges have length 1.

When the coloring has no red ℓ_2, every pair term at distance one vanishes, and the remaining coefficients and indicators are integers, so the local computation reduces to a simple integer calculation.

Lemma 20. *If a coloring of P has no red ℓ_2 and none of the six sets $P_{i,s}$ is entirely blue, then $\mathcal{L} \geq 0$.*

Proof. Under the vertex ordering $00, 01, \ldots, 08, 10, 11, \ldots, 18$, the 33 pairs $\{p, q\} \subset P$ with $|v_p - v_q| = 1$ are

$$\begin{aligned} \{v_{ij}, v_{i,j+1}\} &\quad (i \in \{0,1\},\ 0 \leq j \leq 7), \\ \{v_{0j}, v_{1j}\} &\quad (0 \leq j \leq 8), \\ \{v_{0j}, v_{1,j-1}\} &\quad (1 \leq j \leq 8). \end{aligned}$$

Of the 2^{18} subsets $R \subseteq P$, 751 of these contain none of the 33 pairs and meet all six sets $P_{i,s}$. On each of the 751 subsets, we can check that the integer expression (3.1) is at least 0 (Appendix A.2). □

Suppose a coloring of $\mathbb{E}^2$ has no red ℓ_2 and no blue ℓ_7. Let μ be the G_2-invariant measure from Lemma 10 and r the red indicator at 0; then $\mathbb{E}_\mu[\mathcal{L}] \geq 0$ by Lemma 20. Since $|e_1| = |e_2| = 1$ and $e_1 \cdot e_2 = 1/2$, any two labels $p = (i, j)$ and $q = (i', j')$ satisfy $|v_p - v_q|^2 = Q(p - q)$ with $Q(a, b) = a^2 + ab + b^2$; summing a_{pq} over each value of $Q(p - q)$ (Table 4 in Appendix A.4) defines the kernel

$$\begin{aligned} K_7(t) = &-319 - 489J_0(t) + 1035J_0(\sqrt{3}t) - 669J_0(2t) \\ &+ 600J_0(\sqrt{7}t) - 700J_0(3t) + 548J_0(\sqrt{13}t) \\ &- 355J_0(4t) + 441J_0(\sqrt{21}t) - 105J_0(5t) \\ &+ 339J_0(\sqrt{31}t) - 114J_0(6t) - 139J_0(\sqrt{43}t) \\ &+ 16J_0(7t) - 90J_0(\sqrt{57}t). \end{aligned} \tag{3.2}$$

Proposition 21. *There is a nonnegative Borel measure $\bar{\nu}_r$ on $[0, \infty]$ of mass $\mathbb{E}_\mu[r]$ satisfying*

$$0 \leq \mathbb{E}_\mu[\mathcal{L}] = -1 + \int_{[0,\infty]} K_7(t)\, d\bar{\nu}_r(t). \tag{3.3}$$

Proof. We have $\sum_{p \in P} c_p = -319$ and $\mathbb{E}_\mu[r_p r_q] = C_r(v_q - v_p)$. Applying (2.2) to $h = r$ (using $\mathbb{E}_\mu[r^2] = \mathbb{E}_\mu[r]$ and $J_0(0) = 1$) gives (3.3). □

Proposition 22. *For all $t \geq 0$, $K_7(t) < 0$.*

Proof. See Appendix A.4: adaptive stepping with $|K_7'| \leq 11682.22$ verifies $K_7(t) < 0$ on $[0, 100]$, while Lemma 44(iv) gives $K_7(t) < -319 + 3000/\sqrt{t} \leq -19$ for $t \geq 100$. □

Proof of Theorem 1. Because $K_7 \le 0$ on $[0,\infty]$ (with $K_7(\infty) = -319$), Propositions 21 and 22 give $0 \le \mathbb{E}_\mu[\mathcal{L}] \le -1$, which is absurd. □

4. Three-colorings of the plane

With $e_1 = (1,0)$ and $e_2 = (1/2, \sqrt{3}/2)$, put $v_{ij} = (i-1)e_1 + (j-1)e_2$ $(i,j \in \{0,1,2\})$ and form the nine-point triangular grid $\Lambda = \{v_{ij} : i,j \in \{0,1,2\}\}$ of Figure 2, abbreviated by ij for v_{ij}.

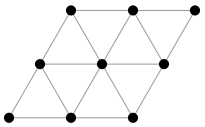


Figure 2. The nine-point triangular-lattice configuration Λ.

Using the same squared-distance form $Q(a,b) = a^2 + ab + b^2$, we have $|v_{ij} - v_{i'j'}|^2 = Q((i,j)-(i',j'))$. The nonzero squared distances in Λ are $1, 3, 4, 7, 12$. The configurations isometric to ℓ_3 used below lie in the three unit directions $(1,0)$, $(0,1)$, and $(1,-1)$.

The four-element group $\mathcal{G} = \langle \sigma, \tau \rangle$ generated by $\sigma(i,j) = (j,i)$ and $\tau(i,j) = (2-i, 2-j)$ acts isometrically on Λ, partitioning the 9 vertices into 4 orbits $\mathcal{O}_V$,

representative	orbit	size
00	$\{00, 22\}$	2
01	$\{01, 10, 12, 21\}$	4
02	$\{02, 20\}$	2
11	$\{11\}$	1

and the $\binom{9}{2} = 36$ unordered pairs into 13 orbits $\mathcal{O}_P$. For $U \in \mathcal{O}_P$, write $q(U) = |v-w|^2$ $(\{v,w\} \in U)$. Given $c : \Lambda \to \{G,R,B\}$, set $g = \mathbf{1}_G$ and $s = \mathbf{1}_R - \mathbf{1}_B$ (so $g^2 = g$, $s^2 = 1-g$, $gs = 0$), and define the orbit averages $[g]_O = |O|^{-1} \sum_{v \in O} g(v)$ $(O \in \mathcal{O}_V)$ and

$$[gg]_U = \frac{1}{|U|} \sum_{\{v,w\} \in U} g(v)g(w), \qquad [ss]_U = \frac{1}{|U|} \sum_{\{v,w\} \in U} s(v)s(w) \qquad (U \in \mathcal{O}_P).$$

With the integer coefficients c_O, a_U, b_U listed in Tables 12 and 13, set

$$\mathcal{L}(c) = -196 + \sum_{O \in \mathcal{O}_V} c_O [g]_O + \sum_{U \in \mathcal{O}_P} \Big(a_U [gg]_U + b_U [ss]_U \Big).$$

On the index set $V = \{0,1,2\}^2$, let

$$\mathcal{E}_1 = \{\{p,q\} \subseteq V : Q(p-q) = 1\},$$
$$\mathcal{I}_3 = \{(p, p+u, p+2u) : u \in \{(1,0),(0,1),(1,-1)\},\ p, p+u, p+2u \in V\},$$

and let $\mathcal{A}_3$ be the set of three-colorings $c : V \to \{G,R,B\}$ having no monochromatic red or blue pair in $\mathcal{E}_1$ and no all-green triple in $\mathcal{I}_3$.

Lemma 23. *Every coloring $c \in \mathcal{A}_3$ of Λ with no red ℓ_2, no blue ℓ_2, and no green ℓ_3 satisfies $\mathcal{L}(c) \ge 0$.*

Proof. Since $|O|$ and $|U|$ divide 4, the combination

$$\begin{aligned} 4\mathcal{L}(c) = &-784 + \sum_{O\in\mathcal{O}_V} \frac{4c_O}{|O|} \sum_{v\in O} g(v) \\ &+ \sum_{U\in\mathcal{O}_P} \frac{4}{|U|} \left(a_U \sum_{\{v,w\}\in U} g(v)g(w) + b_U \sum_{\{v,w\}\in U} s(v)s(w) \right) \end{aligned}$$

is an integer for every c; checking all $|\mathcal{A}_3| = 304$ colorings in $\mathcal{A}_3$ (Appendix A.2) gives $4\mathcal{L}(c) \geq 0$. □

Suppose a coloring of $\mathbb{E}^2$ has no red ℓ_2, no blue ℓ_2, and no green ℓ_3, and let μ be the G_2-invariant measure from Lemma 10 with $g = \mathbf{1}_G(0)$ and $s = \mathbf{1}_R(0) - \mathbf{1}_B(0)$, so $\mathbb{E}_\mu[\mathcal{L}] \geq 0$. Collecting the coefficients in Tables 12–13 by $q(U) \in \{1, 3, 4, 7, 12\}$ gives the kernels

$$\begin{aligned} H_g(t) = &-603 - 1050J_0(t) + 1610J_0(\sqrt{3}t) - 1152J_0(2t) \\ &+ 1000J_0(\sqrt{7}t) + 375J_0(\sqrt{12}t), \end{aligned} \tag{4.1}$$

$$\begin{aligned} H_s(t) = &-442J_0(t) + 138J_0(\sqrt{3}t) - 12J_0(2t) \\ &+ 300J_0(\sqrt{7}t) + 200J_0(\sqrt{12}t). \end{aligned} \tag{4.2}$$

Proposition 24. *There are nonnegative measures $\bar{\nu}_g, \bar{\nu}_s$ on $[0,\infty]$ with $\bar{\nu}_g([0,\infty]) + \bar{\nu}_s([0,\infty]) = 1$ and*

$$0 \leq \mathbb{E}_\mu[\mathcal{L}] = -196 + \int_{[0,\infty]} H_g \, d\bar{\nu}_g + \int_{[0,\infty]} H_s \, d\bar{\nu}_s. \tag{4.3}$$

Proof. We have $\sum_{O\in\mathcal{O}_V} c_O = -603$ and $g^2 = g$, so applying (2.2) to g and s gives (4.3), and $\bar{\nu}_g([0,\infty]) + \bar{\nu}_s([0,\infty]) = \mathbb{E}_\mu[g^2 + s^2] = 1$. □

Proposition 25. *For all $t \geq 0$, $H_g(t) < 193$ and $H_s(t) < 193$.*

Proof. Table 5 in Appendix A.5 bounds $H_g \leq 192.69$ and $H_s \leq 186.84$ on $[0, 25]$ using grid step $1/400$, and Lemma 44(iii) gives $H_g(t) < 55$ and $H_s(t) < 143$ for $t \geq 25$. □

Proof of Theorem 2. By Proposition 25, as well as $H_g(\infty) = -603$ and $H_s(\infty) = 0$, we can show that $H_g, H_s \leq 193$ on $[0,\infty]$. Since $\bar{\nu}_g([0,\infty]) + \bar{\nu}_s([0,\infty]) = 1$, we can apply Proposition 24 to get $0 \leq \mathbb{E}_\mu[\mathcal{L}] \leq -196 + 193 = -3$, contradiction. □

5. Lines in higher dimensions

For a red–blue coloring, put $r = \mathbf{1}_R$ and $s = \mathbf{1}_R - \mathbf{1}_B = 2r - 1$. On a line $x + \mathbb{R}\omega$ with $\omega \in \mathbb{S}^{d-1}$, write $r_j = r(x + j\omega)$ and $s_j = s(x + j\omega)$. Avoiding a red ℓ_a and a blue ℓ_b simply means the binary sequence $(r_j)_{j\in\mathbb{Z}}$ can never contain 1^a or 0^b as a run of consecutive entries (and in the symmetric case $a = b = 4$, we never see four identical signs s_j in a row). To handle Theorems 3–6 all at once, we index the four problems by $q \in Q = \{1, 2, 3, 4\}$ and record the ambient dimension d_q, the set of forbidden blocks F_q, and the step length m_q in the table below:

q	relation	d_q	F_q	m_q
1	(ℓ_3, ℓ_4)	3	$\{111, 0000\}$	3
2	(ℓ_3, ℓ_5)	4	$\{111, 00000\}$	6
3	(ℓ_3, ℓ_6)	7	$\{111, 000000\}$	6
4	(ℓ_4, ℓ_4)	4	$\{1111, 0000\}$	4

Throughout what follows, we say a binary word avoids F_q whenever none of the strings in F_q appear inside it as a contiguous block. Define the de Bruijn vertex and edge sets (cf. Table 3) by $\mathcal{V}_q = \{u \in \{0,1\}^{m_q} : u \text{ avoids } F_q\}$ and

$$\mathcal{E}_q = \Big\{(u,v) \in \mathcal{V}_q^2 : u_1 \cdots u_{m_q-1} = v_0 \cdots v_{m_q-2} \text{ and } u_0 \cdots u_{m_q-1} v_{m_q-1} \text{ avoids } F_q\Big\}.$$

On the (m_q+1)-bit word $u_0 \cdots u_{m_q-1} v_{m_q-1}$ of an edge $(u,v) \in \mathcal{E}_q$, put

$$\begin{aligned}
w_1 &= 4r_0 - 1 - 3r_0r_1 - 2r_0r_2,\\
w_2 &= -6 + 28r_0 - 20r_0r_1 - 14r_0r_2 - 3r_0r_3 - 4r_0r_4 + 2r_0r_5 + 2r_0r_6,\\
w_3 &= -2 + 11r_0 - 9r_0r_1 - 5r_0r_2 - 2r_0r_4 - r_0r_5 + r_0r_6,\\
w_4 &= -1 - 6s_0s_1 - 4s_0s_2 - 2s_0s_3 + s_0s_4,
\end{aligned}$$

and take the integer potentials $\phi_q : \mathcal{V}_q \to \mathbb{Z}$ from Tables 18–21 (Appendix B).

Proposition 26. *For $q \in Q$ and $(u,v) \in \mathcal{E}_q$, $w_q(u,v) + \phi_q(u) - \phi_q(v) \geq 0$.*

Proof. We look at the coboundary difference $\Delta_q(u,v) = w_q(u,v) + \phi_q(u) - \phi_q(v)$ on each directed edge. Take, for instance, the edge $(000, 001) \in \mathcal{E}_1$ corresponding to the four-letter word 0001: here $w_1(0001) = -1$, and reading off $\phi_1(000) = 0$ and $\phi_1(001) = -1$ from Table 18 gives $\Delta_1(000, 001) = -1 + 0 - (-1) = 0$. Across $q = 1, 2, 3, 4$, the graphs $\mathcal{E}_q$ have 12, 73, 78, and 26 edges, respectively. Once we plug in the integer potentials from Appendix B, a quick check shows $\Delta_q(u,v) \geq 0$ across all edges (we record the full verification in Appendix A.2). □

Next, take a G_{d_q}-invariant probability measure μ on colorings that avoid the monochromatic lines from row q. Whenever we fix a unit vector $\omega \in \mathbb{F}^{d_q}$, the colors at the points $0, \omega, \ldots, m_q\omega$ give us a valid directed edge $(u,v) \in \mathcal{E}_q$. Because v is just a shift of u by ω, translation invariance gives $\mathbb{E}_\mu[\phi_q(u)] = \mathbb{E}_\mu[\phi_q(v)]$; telescoping the coboundary difference via Proposition 26 then forces $\mathbb{E}_\mu[w_q] \geq 0$. For the asymmetric cases $q \in \{1,2,3\}$, we write $p = \mathbb{E}_\mu[r]$ so that $\mathbb{E}_\mu[rT_{k\omega}r] = p^2 + C_{r-p}(k\omega)$ with variance $\mathbb{E}_\mu[(r-p)^2] = p(1-p)$, whereas for $q = 4$ we simply use $s^2 = 1$. In Theorem 19, the resulting spherical kernels are

$$\begin{aligned}
H_1(t) &= -3\Omega_3(t) - 2\Omega_3(2t),\\
H_2(t) &= -20\Omega_4(t) - 14\Omega_4(2t) - 3\Omega_4(3t)\\
&\qquad - 4\Omega_4(4t) + 2\Omega_4(5t) + 2\Omega_4(6t),\\
H_3(t) &= -9\Omega_7(t) - 5\Omega_7(2t) - 2\Omega_7(4t) - \Omega_7(5t) + \Omega_7(6t),\\
H_4(t) &= -6\Omega_4(t) - 4\Omega_4(2t) - 2\Omega_4(3t) + \Omega_4(4t).
\end{aligned}$$

Proposition 27. *For all $t \geq 0$, we have $H_1(t) < 0.7$, $H_2(t) < 2.95$, $H_3(t) < 0.4$, and $H_4(t) < 0.9$.*

Proof. Table 6 bounds each kernel on a finite interval. The tail bounds in Table 7 cover the remaining values of t. Both verifications appear in Appendix A.6. □

Proofs of Theorems 3–6. Assume that a coloring avoids the monochromatic lines in the theorem under consideration, and let μ be the corresponding invariant measure. Theorem 19 gives the four inequalities

$$\begin{aligned} 0 &\leq -1 + 4p - 5p^2 + \int H_1 \, d\eta_1, \\ 0 &\leq -6 + 28p - 37p^2 + \int H_2 \, d\eta_2, \\ 0 &\leq -2 + 11p - 16p^2 + \int H_3 \, d\eta_3, \\ 0 &\leq -1 + \int H_4 \, d\nu_4. \end{aligned}$$

The first three measures have mass $p(1-p)$, while the fourth has mass one. Using the kernel bounds, which also hold at ∞ since $H_q(\infty) = 0$, we get the quadratic upper bounds as presented in Table 2 for the first three cases. All three quadratics are negative for every real p, since their leading coefficients and discriminants are negative. In the fourth case, the upper bound is $-1 + 0.9 = -0.1$. Thus all four cases are impossible.

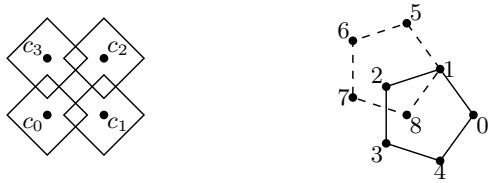
q	Upper bound	Discriminant
1	$-1 + 4.7p - 5.7p^2$	-0.71
2	$-6 + 30.95p - 39.95p^2$	-0.8975
3	$-2 + 11.4p - 16.4p^2$	-1.24

TABLE 2. Polynomial bounds for the first three line inequalities.

□

6. The centered unit square W_4

Figure 3 displays the twenty-point configuration of four copies of W_4 used to prove Theorem 7, together with the nine-point configuration of two copies of P_5 used in Section 7.



(*a*) W_4 (*b*) P_5

FIGURE 3. The planar configurations used for W_4 and P_5.

6.1. **A bound on the red probability.** For a G_2-invariant probability measure μ on $\mathcal{X}$, let $\mathcal{H}_c \subset L^2(\mu)$ be the subspace of functions f such that $\|T_z f - f\|_{L^2(\mu)} \to 0$ as $|z| \to 0$ ($z \in \mathbb{F}^2$), and write λ_f for the Bochner measure of $z \mapsto \mathbb{E}_\mu[\overline{f} T_z f]$ on $D = \widehat{\mathbb{F}^2}$. By [29, Lemma 4.1 and Proposition 4.2], $\mathcal{H}_c$ is the L^2-space of a G_2-invariant factor σ-algebra $\mathcal{A}$ and admits jointly measurable $\mathbb{R}^2$-translates:

Theorem 28. *There is a G_2-invariant sub-σ-algebra $\mathcal{A}$ such that $\mathcal{H}_c = L^2(\mathcal{X}, \mathcal{A}, \mu)$, with $f \in \mathcal{H}_c$ if and only if $\lambda_f(D \setminus C) = 0$, and $f \in \mathcal{H}_c^\perp$ if and only if $\lambda_f(C) = 0$; orthogonal projection onto $\mathcal{H}_c$ coincides with conditional expectation $\mathbb{E}_\mu[\cdot \mid \mathcal{A}]$.*

Proposition 29. *By L^2 continuity, the translation action of $\mathbb{F}^2$ on $\mathcal{H}_c$ extends, and for every $q \in \mathcal{H}_c$ with $0 \le q \le 1$, there exists a jointly measurable function $H : \mathbb{R}^2 \times \mathcal{X} \to [0,1]$ such that $H(x, \cdot) = T_x q$ in $L^2(\mu)$ for almost every $x \in \mathbb{R}^2$.*

By Ambrus et al. [1, Theorem 1], any measurable set $A \subset \mathbb{R}^2$ containing no unit-distance pair ℓ_2 has upper density at most $0.2470 < 1/4$; writing $B_T = \{x \in \mathbb{R}^2 : |x| \le T\}$, we use only the bound $1/4$:

Theorem 30. *Every measurable set $A \subset \mathbb{R}^2$ with no ℓ_2 satisfies $\limsup_{T\to\infty} |A \cap B_T|/|B_T| \le 1/4$.*

Following the density-point argument of Falconer [20] and Payne [30] (cf. [29, Lemma 4.3]), call $x \in \mathbb{R}^2$ a density-one point of a measurable set $A \subset \mathbb{R}^2$ if $|A \cap B(x,\varepsilon)|/(\pi\varepsilon^2) \to 1$ as $\varepsilon \to 0$, where $B(x,\varepsilon)$ is the disk of radius ε about x.

Lemma 31. *Let $A \subset \mathbb{R}^2$ be measurable with $\mathbf{1}_A(x)\mathbf{1}_A(x+u) = 0$ for almost every $(x,u) \in \mathbb{R}^2 \times \mathbb{S}^1$ under $\mathrm{d}x\,\mathrm{d}\sigma(u)$. Then the set A^* of density-one points of A is measurable, has $|A \triangle A^*| = 0$, and contains no ℓ_2.*

Proof. The Lebesgue density theorem gives measurability of A^* and $|A \triangle A^*| = 0$. Suppose that $x_0, y_0 \in A^*$ are distance one apart. For $0 < \varepsilon < 1/2$, consider $(x, x+u)$ such that $x \in B(x_0,\varepsilon)$, $u \in \mathbb{S}^1$, and $x + u \in B(y_0,\varepsilon)$. Finally, let ν_ε be the image of $\mathrm{d}x\,\mathrm{d}\sigma(u)$ on this set under $(x,u) \mapsto (x, x+u)$.

For $x \in B(x_0, \varepsilon/2)$, an arc of length at least $c_0\varepsilon$ of the circle $x + \mathbb{S}^1$ lies in $B(y_0,\varepsilon)$, so consequently

$$\nu_\varepsilon\Big(B(x_0,\varepsilon) \times B(y_0,\varepsilon)\Big) \ge c\varepsilon^3$$

for an absolute constant $c > 0$. Additionally, each marginal of ν_ε has density at most $C\varepsilon$ with respect to planar Lebesgue measure, since for a fixed first or second coordinate, the admissible directions occupy an arc of length at most $C_0\varepsilon$.

Both x_0 and y_0 are density-one points of A. For sufficiently small ε,

$$|B(x_0,\varepsilon) \setminus A| + |B(y_0,\varepsilon) \setminus A| < \frac{c\varepsilon^2}{2C}.$$

We can now use the marginal bounds to show that $\nu_\varepsilon(A \times A) > 0$, which contradicts our assumption of almost-everywhere vanishing of $\mathbf{1}_A(x)\mathbf{1}_A(x+u)$. □

Proposition 32. *Let μ be a G_2-invariant probability measure on colorings of $\mathbb{F}^2$ with no red ℓ_2. If r denotes the red indicator at 0, then $\mathbb{E}_\mu[r] \le 1/4$.*

Proof. Let $p = \mathbb{E}_\mu[r]$ and $q = \mathbb{E}_\mu[r \mid \mathcal{A}] \in \mathcal{H}_c$, as in Theorem 28, so that $0 \le q \le 1$, $\mathbb{E}_\mu[q] = p$, and $r - q \in \mathcal{H}_c^\perp$. Both q and $r - q$ are K-invariant. Since the Bochner measure of $r - q$ has $\lambda_{r-q}(C) = 0$, Theorem 12 gives $\lambda_{r-q} = am_D$ for some $a \ge 0$; therefore $\mathbb{E}_\mu[(r-q)T_z(r-q)] = 0$ for $z \in \mathbb{F}^2 \setminus \{0\}$. Translations preserve $\mathcal{H}_c$ and its orthogonal complement, and $\mathbb{E}_\mu[rT_z r] = 0$ at unit distance. Hence

$$0 = \mathbb{E}_\mu[rT_z r] = \mathbb{E}_\mu[qT_z q] + \mathbb{E}_\mu[(r-q)T_z(r-q)] = \mathbb{E}_\mu[qT_z q] \qquad (z \in \mathbb{F}^2,\ |z| = 1).$$

It's now clear that by L^2-continuity, this identity holds with z replaced by every $u \in \mathbb{S}^1$.

Define $H : \mathbb{R}^2 \times \mathcal{X} \to [0,1]$ as the jointly measurable representative given by Proposition 29, and notice that $\mathbb{E}_\mu[H(x,\cdot)] = p$ for almost every x, and by Fubini's theorem, $\mathbb{E}_\mu[H(x,\cdot)H(x+u,\cdot)] = \mathbb{E}_\mu[qT_u q] = 0$ for almost every $(x,u) \in \mathbb{R}^2 \times \mathbb{S}^1$. Now by Tonelli's theorem, for μ-almost every ω the set $A_\omega = \{x \in \mathbb{R}^2 : H(x,\omega) > 0\}$ satisfies Lemma 31, since its density-one representative A_ω^* contains no ℓ_2, and $|A_\omega \triangle A_\omega^*| = 0$. As $H(\cdot,\omega) \le \mathbf{1}_{A_\omega}$, the bound applies to its ball averages.

For integers $n \ge 1$, set $Z_n(\omega) = |B_n|^{-1} \int_{B_n} H(x,\omega)\,dx$. Since $H(\cdot,\omega) \le \mathbf{1}_{A_\omega}$, Theorem 30 implies $\limsup_n Z_n(\omega) \le 1/4$ almost surely. On the other hand, we have $0 \le Z_n \le 1$ and $\mathbb{E}_\mu[Z_n] = p$. Now we apply Fatou's lemma to $1 - Z_n$, which gives:

$$1 - p \ge \mathbb{E}_\mu\Big[\liminf_{n\to\infty}(1 - Z_n)\Big] = 1 - \mathbb{E}_\mu\Big[\limsup_{n\to\infty} Z_n\Big] \ge \frac{3}{4}$$

and therefore $p \le 1/4$. □

6.2. **The local inequality for W_4.** Let $e_1 = (1,0)$, $e_2 = (0,1)$, and

$$(u_0, u_1, u_2, u_3, u_4) = \frac{1}{\sqrt{2}}(0, e_1, e_2, -e_1, -e_2).$$

For the unit-square vertices $c_0 = (0,0)$, $c_1 = (1,0)$, $c_2 = (1,1)$, and $c_3 = (0,1)$, set $v_{5k+j} = c_k + u_j$ for $0 \le k \le 3$ and $0 \le j \le 4$ (Figure 3(a)). The four copies of W_4 have index sets $I_k = \{5k, \dots, 5k+4\}$. The 36 unit-distance pairs are

$$\mathcal{E}_4 = \{(i,j) : 0 \le i < j \le 19,\ |v_i - v_j| = 1\},$$

as listed in Appendix B.1. If $r_i \in \{0,1\}$ is the red indicator at v_i, any (ℓ_2, W_4)-free coloring of $\{v_0, \dots, v_{19}\}$ satisfies $r_i r_j = 0$ for $(i,j) \in \mathcal{E}_4$, $\sum_{i \in I_k} r_i \ge 1$ for $0 \le k \le 3$, and we set

$$\mathcal{L}_{W_4}(r) = c + \sum_{i=0}^{19} \alpha_i r_i + \sum_{0 \le i < j \le 19} \beta_{ij} r_i r_j \qquad \Big(r \in \{0,1\}^{20}\Big),$$

with $c = -24.994229$ and $10^6\alpha_i, 10^6\beta_{ij} \in \mathbb{Z}$ from Tables 9 and 10.

Lemma 33. *Every $r \in \{0,1\}^{20}$ with $r_i r_j = 0$ on $\mathcal{E}_4$ and $\sum_{i \in I_k} r_i \ge 1$ $(0 \le k \le 3)$ satisfies $\mathcal{L}_{W_4}(r) \ge 0$.*

Proof. We can check all 2338 admissible vectors $r \in \{0,1\}^{20}$ to find that $10^6\mathcal{L}_{W_4}(r) \ge 0$ (see Appendix A.2). □

Let $K_{W_4}(t) = \sum_{i<j} \beta_{ij} J_0(|v_i - v_j|t)$ (using Table 11). Under a G_2-invariant measure μ on (ℓ_2, W_4)-free colorings with $p = \mathbb{E}_\mu[r]$, we have $C_r(z) = 0$ for $|z| = 1$, so Corollary 14 implies that $\int_{[0,\infty]} J_0(t)\, d\bar{\eta}(t) = -p^2$ and

$$\int_{[0,\infty]} K_{W_4}(t)\, d\bar{\eta}(t) = \int_{[0,\infty]} \Big(K_{W_4}(t) - 600 J_0(t)\Big)\, d\bar{\eta}(t) - 600p^2.$$

Proposition 34. *For $t \geq 0$, $K_{W_4}(t) - 600J_0(t) < 46$.*

Proof. The calculation in Appendix A.3 gives the bound on $[0, 125]$, where we use Taylor estimates on 2500 subintervals of length $1/20$. For $t \geq 125$, Lemma 44(ii) gives a stronger estimate $|K_{W_4}(t) - 600J_0(t)| < 40$. □

Proof of Theorem 7. Let μ be the G_2-invariant measure obtained from an (ℓ_2, W_4)-free coloring. Proposition 32 gives $p = \mathbb{E}_\mu[r] \leq 1/4$, and Corollary 14 gives $\bar{\eta}([0,\infty]) = p(1-p)$.

The coefficient sums are $A = \sum_i \alpha_i = 139.366276$ and $B = \sum_{i<j} \beta_{ij} = 303.205864$. Use the local inequality, together with the estimate $K_{W_4} - 600J_0 \leq 46$ (also at ∞), to obtain

$$0 \leq \mathbb{E}_\mu[\mathcal{L}_{W_4}] \leq c + (A+46)p + (B-646)p^2 \leq -24.99 + 185.37p - 342.79p^2 =: \Phi(p).$$

But $\Phi'(p) = 185.37 - 685.58p > 13$ on $[0, 1/4]$, and

$$\boxed{\Phi(1/4) = -0.071875}.$$

This forces $\Phi(p) < 0$, which directly contradicts our first inequality. □

7. The regular pentagon

To show that $\mathbb{E}^2 \to (\ell_2, P_5)$, we consider a regular unit pentagon $B_0 = \{u_0, \ldots, u_4\}$ with circumradius $R_5 = 1/(2\sin(\pi/5))$ and vertices given explicitly by

$$u_j = \frac{1}{2\sin(\pi/5)} \left(\cos\frac{2\pi j}{5}, \sin\frac{2\pi j}{5}\right)$$

for $0 \leq j \leq 4$.

Then, we translate B_0 by the edge vector $s = u_1 - u_0$ to form a second pentagon $B_1 = B_0 + s$. By construction, B_0 and B_1 meet at $u_1 = u_0 + s$, and we claim they share no other vertices. To show this, note that any second overlap would require $u_i - u_j = s$. A unit chord of the regular pentagon is a side, and its ten directed side vectors are distinct; hence $u_i - u_j = u_1 - u_0$ only for $(i, j) = (1, 0)$, with indices modulo 5.

Thus $B_0 \cup B_1$ has nine points, which we will label by $v_i = u_i$ for $0 \leq i \leq 4$ and $v_{4+j} = u_j + s$ for $1 \leq j \leq 4$. It is now clear that two pentagons have index sets $I_0 = \{0, 1, 2, 3, 4\}$ and $I_1 = \{1, 5, 6, 7, 8\}$, and there are thirteen unit-distance pairs among these points. We will write $\mathcal{E}_5 = \{(i, j) : i < j,\ |v_i - v_j| = 1\}$; Appendix B.4 lists the pairs and their distances.

Now, one can see that any (ℓ_2, P_5)-free coloring of $\{v_0, \ldots, v_8\}$ satisfies $r_i r_j = 0$ for $(i, j) \in \mathcal{E}_5$, $\sum_{i\in I_0} r_i \geq 1$, and $\sum_{i\in I_1} r_i \geq 1$, where r_i is the red indicator at v_i. For

$r \in \{0,1\}^9$, define

$$\mathcal{L}_{P_5}(r) = -77 + \sum_{i=0}^{8} \alpha_i r_i + \sum_{0 \le i < j \le 8} \beta_{ij} r_i r_j.$$

The coefficients are listed in Table 17 of Appendix B.4.

Lemma 35. *If $r \in \{0,1\}^9$ satisfies $r_i r_j = 0$ for all $(i,j) \in \mathcal{E}_5$ and $\sum_{i \in I_0} r_i \ge 1$, $\sum_{i \in I_1} r_i \ge 1$, then $\mathcal{L}_{P_5}(r) \ge 0$.*

Proof. Of the $2^9 = 512$ vectors $r \in \{0,1\}^9$, 42 satisfy these constraints, and on each of those 42 vectors, $\mathcal{L}_{P_5}(r)$ is at least 0 (see Appendix A.2). □

Collecting the pair coefficients β_{ij} by $|v_i - v_j|^2$ gives

$$\begin{aligned} K_{P_5}(t) = 200 J_0\left(\sqrt{\frac{3-\sqrt{5}}{2}}\, t\right) - 1048 J_0(t) \\ + 100 J_0\left(\sqrt{\frac{5-\sqrt{5}}{2}}\, t\right) - 385 J_0\left(\sqrt{\frac{3+\sqrt{5}}{2}}\, t\right). \end{aligned}$$

Proposition 36. *For all $t \ge 0$, $K_{P_5}(t) < 393$.*

Proof. In Appendix A.5 (Table 5), evaluation at the 2400 cell midpoints of $[0,12]$ with $|K'_{P_5}| \le 1330$ gives $K_{P_5}(t) \le 392.33$ on $[0,12]$, and Lemma 44(ii) gives $|K_{P_5}(t)| < 392$ for $t \ge 12$. □

Proof of Theorem 8. For an (ℓ_2, P_5)-free coloring with G_2-invariant measure μ and $p = \mathbb{E}_\mu[r]$, we have $\sum_i \alpha_i = 285$, $\sum_{i<j} \beta_{ij} = -1133$, and $K_{P_5} \le 393$ on $[0,\infty]$ ($K_{P_5}(\infty) = 0$). Since $\bar{\eta}([0,\infty]) = p(1-p)$, Lemma 35 and Corollary 14 give

$$\begin{aligned} 0 \le \mathbb{E}_\mu[\mathcal{L}_{P_5}] &= -77 + 285p - 1133p^2 + \int_{[0,\infty]} K_{P_5}\, d\bar{\eta} \\ &\le -77 + 678p - 1526p^2 \\ &= -1 - (1526p^2 - 678p + 76) \le -1, \end{aligned}$$

which is impossible. □

8. Unit squares and tetrahedra in $\mathbb{E}^3$

Put $e = (1,0,0) \in \mathbb{R}^3$ and, for $r \in \{0,1,2\}$, let $v_r = \big(0, \cos(r\pi/3), \sin(r\pi/3)\big)$ and $w_r = \big(0, -\sin(r\pi/3), \cos(r\pi/3)\big)$. For each r, (e, v_r, w_r) is an orthonormal basis of $\mathbb{R}^3$, so $Q_r = \{bv_r + cw_r : b,c \in \{0,1\}\}$ is a unit square in $e^\perp$ and $C_r = Q_r \cup (e + Q_r)$ is a unit cube. Since the six unit vectors v_r, w_r and the three vectors $v_r + w_r$ of norm $\sqrt{2}$ have nine distinct directions in $e^\perp$, we have $Q_r \cap Q_s = \{0\}$ and $C_r \cap C_s = \{0, e\}$ ($r \ne s$), so $\mathcal{P} = C_0 \cup C_1 \cup C_2$ has $|\mathcal{P}| = 2 + 3 \cdot 6 = 20$ vertices (Figure 4).

Both $\mathcal{S}$ and $\mathcal{T}$ have pairwise distance multiset

$$\{1, 1, 1, 1, \sqrt{2}, \sqrt{2}\}.$$

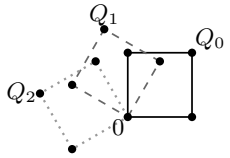


FIGURE 4. The three cross-sections Q_r of the configuration $\mathcal{P}$ in the plane perpendicular to the common edge e. Solid, dashed, and dotted curves correspond to Q_0, Q_1, and Q_2, respectively.

The set $Y = \{0, w_0, e, w_1\} \subset \mathcal{P}$ has this multiset, and

$$\det[w_0, e, w_1] = -\sqrt{3}/2 \neq 0.$$

Thus Y is noncoplanar and isometric to $\mathcal{T}$.

Lemma 37. *Let $X \subset \mathbb{E}^3$ be a four-point set with distances $\{1, 1, 1, 1, \sqrt{2}, \sqrt{2}\}$. If the two $\sqrt{2}$-pairs are disjoint, then $X \cong \mathcal{S}$, and otherwise $X \cong \mathcal{T}$.*

Proof. First, suppose that $\{x_1, x_3\}$ and $\{x_2, x_4\}$ are disjoint $\sqrt{2}$-pairs, and translate the midpoint of x_1x_3 to 0. Then $x_3 = -x_1$, $|x_1|^2 = 1/2$, and $|x_2 \pm x_1| = |x_4 \pm x_1| = 1$, so $x_1 \perp x_2, x_4$ and $|x_2|^2 = |x_4|^2 = 1/2$. Since $|x_2 - x_4|^2 = 2$, we get $x_4 = -x_2$ and $X = \{\pm x_1, \pm x_2\} \cong \mathcal{S}$.

If instead the $\sqrt{2}$-pairs share a vertex, we can translate the vertex belonging to neither pair to 0 and write $X = \{0, a, b, c\}$ with $|a - b| = |b - c| = \sqrt{2}$ and $|a| = |b| = |c| = |a - c| = 1$. Then we can compute the Gram matrix of (a, b, c) to get

$$\begin{pmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1 \end{pmatrix},$$

which is the same as the Gram matrix of the nonzero vertices of $\mathcal{T}$. Therefore $X \cong \mathcal{T}$. □

Consider the two isometries $\tau(x, y, z) = (1 - x, y, z)$ and

$$\sigma(x, y, z) = \left(x, -\frac{\sqrt{3}}{2}y - \frac{1}{2}z, -\frac{1}{2}y + \frac{\sqrt{3}}{2}z\right),$$

and let $G = \langle \tau, \sigma \rangle$. We follow the labels $p_0, \ldots, p_{19}$ from Appendix B.6.

Lemma 38. *The set $\mathcal{P}$ contains 22 copies of $\mathcal{S}$ and 24 copies of $\mathcal{T}$. The group G preserves $\mathcal{P}$, has order 4, and has 56 orbits on its 190 unordered pairs.*

Proof. All points of $\mathcal{P}$ lie in $\mathbb{Q}(\sqrt{3})^3$. Of the $\binom{20}{4} = 4845$ four-element subsets, the calculation in Appendix A.7 finds 46 with squared-distance multiset $\{1, 1, 1, 1, 2, 2\}$. In 22 of these the two pairs at squared distance 2 are disjoint and in the other 24 they meet, where we can use the preceding lemma to identify the two.

Now, we need to count the pair orbits. For $a, b, c \in \{0, 1\}$, the map τ sends $ae + bv_r + cw_r$ to $(1-a)e + bv_r + cw_r$, so it preserves each cube C_r. Furthermore, $\sigma e = e$ and $(\sigma v_r, \sigma w_r) = (w_{2-r}, v_{2-r})$, so σ interchanges C_0 and C_2 and fixes C_1. The two maps are distinct commuting involutions, hence $|G| = 4$. Directly counting, their action on

the 190 pairs has 2 orbits of size 1, 14 of size 2, and 40 of size 4, giving us a total of 56. □

With the 56 pair orbits $\mathcal{O}_0, \dots, \mathcal{O}_{55}$ and integer weights $a_0, \dots, a_{55}$ ordered as in Table 23, set $X_j(s) = \sum_{\{p_u, p_v\} \in \mathcal{O}_j} s(p_u)s(p_v)$ for $s : \mathcal{P} \to \{-1, 1\}$ and

$$\mathcal{L}_S(s) = 10^{-3} \left(-36608 + \sum_{j=0}^{55} a_j X_j(s) \right). \tag{8.1}$$

Lemma 39. *Every $s : \mathcal{P} \to \{-1, 1\}$ that is nonconstant on all 22 copies of $\mathcal{S}$ and 24 copies of $\mathcal{T}$ in $\mathcal{P}$ satisfies $\mathcal{L}_S(s) \geq 0$.*

Proof. Testing all 2^{20} maps $s \in \{-1, 1\}^{\mathcal{P}}$ leaves 47946 valid colorings, and $10^3 \mathcal{L}_S(s) \geq 0$ on each (Appendix A.7). □

For $\{p_u, p_v\} \in \mathcal{O}_j$, put $q_j = |p_u - p_v|^2$, sum $B_q = \sum_{0 \leq j \leq 55 :\, q_j = q} |\mathcal{O}_j| a_j$ (Table 22; $\sum_q |B_q| = 620268$), and set

$$K_S(t) = 10^{-3} \left(-36608 + \sum_q B_q \Omega_3(\sqrt{q}\, t) \right). \tag{8.2}$$

Proposition 40. *If a two-coloring of $\mathbb{E}^3$ avoids monochromatic copies of $\mathcal{S}$ and $\mathcal{T}$, then $\int_{[0,\infty]} K_S(t)\, d\nu(t) \geq 0$ for a Borel probability measure ν on $[0, \infty]$.*

Proof. Apply Theorem 19 to $h = s = \mathbf{1}_R(0) - \mathbf{1}_B(0)$ ($s^2 = 1$) under the G_3-invariant measure μ of Lemma 10, using $\mathcal{L}_S \geq 0$ on $R\mathcal{P}$ ($R \in \mathrm{SO}(3, \mathbb{F})$) from Lemma 39. □

Proposition 41. *For all $t \geq 0$, $K_S(t) < -1/2$.*

Proof. Appendix A.7 checks $K_S(t) < -0.5225$ on $[0, 40]$ via the grid $t_j = j/1000$ ($|K_S'| < 931$), and bounds $K_S(t) \leq -36.608 + 1240.536/t < -5.59$ for $t \geq 40$ via $|\Omega_3(\sqrt{q}\, t)| \leq 2/t$. □

Proof of Theorem 9. Because $K_S(t) \leq -1/2$ for all $t \in [0, \infty]$ (with $K_S(\infty) = -36.608$), Proposition 40 gives $0 \leq \int_{[0,\infty]} K_S\, d\nu \leq -1/2$, which is impossible. □

9. Linear programs for the local inequalities

For our planar two-color problems, we shall fix a finite point set $V \subset \mathbb{F}^2$ with unit-distance edges E and target blue subsets $B_1, \dots, B_s \subseteq V$. In this setting, a binary coloring $r = (r_v)_{v \in V} \in \{0, 1\}^V$ is valid whenever $r_u r_v = 0$ for all $\{u, v\} \in E$ and $\sum_{v \in B_i} r_v \geq 1$ for $1 \leq i \leq s$. On such configurations, we study quadratic polynomials

$$\widetilde{\mathcal{L}}(r) = c + \sum_{v \in V} a_v r_v + \sum_{\{u,v\} \subseteq V} b_{uv} r_u r_v$$

with real coefficients c, a_v, and $b_{uv} = b_{vu}$.

Under a G_2-invariant measure μ with $p = \mathbb{E}_\mu[r]$, Corollary 14 gives $C_r(u - v) = p^2 + \int_{[0,\infty]} J_0(|u - v|t)\, d\bar{\eta}(t)$ with $\bar{\eta}([0, \infty]) = p(1 - p)$. Now set $\widehat{\eta} = \bar{\eta}/(p(1 - p))$ for

$0 < p < 1$, and we have $\mathbb{E}_\mu[\widetilde{\mathcal{L}}] = \int_{[0,\infty]} \widetilde{H}(p,t)\, d\widehat{\eta}(t)$, where

$$\widetilde{H}(p,t) = c + p\sum_{v\in V} a_v + \sum_{\{u,v\}\subseteq V} b_{uv}\Big(p^2 + p(1-p)J_0(|u-v|t)\Big).$$

When $p = 0$ or $p = 1$, the function $\widetilde{H}(p,t)$ is independent of t, so any probability measure may be used in place of $\widehat{\eta}$.

For W_4, the unadjusted kernel $\widetilde{H}(p,t)$ is positive at $(p,t) = (1/4, 0)$ ($\widetilde{H}(1/4,0) = 85.648806$), so the proof in Section 6 instead subtracts $600\int_{[0,\infty]} J_0(t)\, d\bar{\eta}(t) = -600p^2$ and combines the bound $K_{W_4} - 600J_0 < 46$ with $p \le 1/4$ (see Appendix A.3).

For the eighteen-point strip P of Section 3 (with valid coloring set $\mathcal{A}_7 \subset \{0,1\}^P$) and the nine-point grid Λ of Section 4, the linear term is absorbed into the Bessel integral via (2.2). On P, with variables x_v, $y_{uv} = y_{vu}$, and δ, we put

$$\widetilde{\mathcal{L}}_7(r) = -1 + \sum_{v\in P} x_v r_v + \sum_{\{u,v\}\subseteq P} y_{uv} r_u r_v,$$
$$\widetilde{K}_7(t) = \sum_{v\in P} x_v + \sum_{\{u,v\}\subseteq P} y_{uv} J_0(|u-v|t),$$

and solve the LP (with 342 bounds $|x_v|, |y_{uv}| \le 100$ on the 171 coefficients and grid $\mathcal{N}_7 \subset [0,100]$; see Table 8):

$$\begin{array}{ll} \text{maximize} & \delta \\ \text{variables} & \delta,\ (x_v)_{v\in P},\ (y_{uv})_{\{u,v\}\subseteq P} \\ \text{subject to} & \widetilde{\mathcal{L}}_7(r) \ge 0 \quad (r \in \mathcal{A}_7), \\ & \widetilde{K}_7(t) \le -\delta \quad (t \in \mathcal{N}_7), \\ & |x_v|, |y_{uv}| \le 100. \end{array}$$

On the nine-point grid Λ with valid three-colorings $\mathcal{A}_3$ and orbit variables $(x_O)_{O\in\mathcal{O}_V}$, $(y_U)_{U\in\mathcal{O}_P}$, $(z_U)_{U\in\mathcal{O}_P}$, we set

$$\widetilde{\mathcal{L}}_3 = -196 + \sum_{O\in\mathcal{O}_V} x_O[g]_O + \sum_{U\in\mathcal{O}_P}\Big(y_U[gg]_U + z_U[ss]_U\Big)$$

and

$$\widetilde{H}_g(t) = \sum_{O\in\mathcal{O}_V} x_O + \sum_{U\in\mathcal{O}_P} y_U J_0(\sqrt{q(U)}t),$$
$$\widetilde{H}_s(t) = \sum_{U\in\mathcal{O}_P} z_U J_0(\sqrt{q(U)}t),$$

so $\mathbb{E}_\mu[\widetilde{\mathcal{L}}_3] = -196 + \int_{[0,\infty]} \widetilde{H}_g\, d\bar{\nu}_g + \int_{[0,\infty]} \widetilde{H}_s\, d\bar{\nu}_s$ with $\bar{\nu}_g([0,\infty]) + \bar{\nu}_s([0,\infty]) = 1$. Imposing a common upper bound $196 - \delta$ on $\widetilde{H}_g$ and $\widetilde{H}_s$ across $t \in \mathcal{N}_{\mathrm{col}}$ (with all 31 variables unconstrained) gives the LP

$$\begin{array}{ll} \text{maximize} & \delta \\ \text{variables} & \delta,\ (x_O)_{O\in\mathcal{O}_V},\ (y_U)_{U\in\mathcal{O}_P},\ (z_U)_{U\in\mathcal{O}_P} \\ \text{subject to} & \widetilde{\mathcal{L}}_3(c) \ge 0 \quad (c \in \mathcal{A}_3), \\ & \widetilde{H}_g(t) \le 196 - \delta,\ \widetilde{H}_s(t) \le 196 - \delta \quad (t \in \mathcal{N}_{\mathrm{col}}). \end{array}$$

In the four higher-dimensional line problems ($q \in \{1,2,3,4\}$) on the de Bruijn graphs $(\mathcal{V}_q, \mathcal{E}_q)$ of Section 5, write $\widetilde{w}_q(u,v)$ for the edge weight on $(u,v) \in \mathcal{E}_q$,

$$\begin{aligned}
\widetilde{w}_1(r) &= -1 + a_1 r_0 + c_{1,1} r_0 r_1 + c_{1,2} r_0 r_2, \\
\widetilde{w}_2(r) &= -6 + a_2 r_0 + \sum_{k=1}^{6} c_{2,k} r_0 r_k, \\
\widetilde{w}_3(r) &= -2 + a_3 r_0 + \sum_{k \in \{1,2,4,5,6\}} c_{3,k} r_0 r_k, \\
\widetilde{w}_4(s) &= -1 + \sum_{k=1}^{4} c_{4,k} s_0 s_k,
\end{aligned}$$

and $\widetilde{H}_q(t) = \sum_k c_{q,k} \Omega_{d_q}(kt)$ for the spherical kernel. For $q \in \{1,2,3\}$ (with $p = \mathbb{E}_\mu[r]$ and $f = r - p$), Theorem 19 gives the (nonnegative) average

$$\widetilde{L}_q = \alpha_q + \beta_q p + S_q p^2 + \int_{[0,\infty]} \widetilde{H}_q(t)\, d\eta_f(t)$$

where $\eta_f([0,\infty]) = p(1-p)$, $\beta_q = a_q$, $S_q = \sum_k c_{q,k}$, and (α_q, β_q, S_q) takes the rounded values $(-1,4,-5)$, $(-6,28,-37)$, and $(-2,11,-16)$ for $q = 1,2,3$; for $q = 4$, $\widetilde{L}_4 = -1 + \int_{[0,\infty]} \widetilde{H}_4\, d\nu_4$ with $\nu_4([0,\infty]) = 1$. With target thresholds $(\kappa_1, \kappa_2, \kappa_3, \kappa_4) = (0.7, 2.95, 0.4, 0.9)$ in dimensions $(d_1, d_2, d_3, d_4) = (3,4,7,4)$ and a reference vertex $u_q^* \in \mathcal{V}_q$ normalized by $\widetilde{\phi}_q(u_q^*) = 0$, the LP for case q is

$$\begin{array}{ll}
\text{maximize} & \delta_q \\
\text{variables} & \delta_q,\ a_q,\ (c_{q,k})_k,\ (\widetilde{\phi}_q(u))_{u \in \mathcal{V}_q} \\
\text{subject to} & \widetilde{w}_q(u,v) + \widetilde{\phi}_q(u) - \widetilde{\phi}_q(v) \geq 0 \quad ((u,v) \in \mathcal{E}_q), \\
& \widetilde{H}_q(t) \leq \kappa_q - \delta_q \quad (t \in \mathcal{N}_q), \\
& \widetilde{\phi}_q(u_q^*) = 0,\ -100 \leq \delta_q \leq 100,\ |a_q|, |c_{q,k}| \leq 100,\ |\widetilde{\phi}_q(u)| \leq 1000
\end{array}$$

When $q = 4$, the variable a_4 is omitted. Table 8 lists the variable and constraint counts, and Appendix A gives the sample grids $\mathcal{N}_q$. The rounded coefficients used in Section 5 are

q	a_q	$(c_{q,k})_k$
1	4	$(-3,-2)$
2	28	$(-20,-14,-3,-4,2,2)$
3	11	$(-9,-5,-2,-1,1)$ for $k = 1,2,4,5,6$
4	–	$(-6,-4,-2,1)$

and the integer functions ϕ_q are given in Tables 18–21.

For the 20-point configuration $\mathcal{P} \subset \mathbb{F}^3$ of Section 8, let $\mathcal{A}_{\mathcal{S},\mathcal{T}} \subset \{-1,1\}^{\mathcal{P}}$ be the set of sign colorings avoiding monochromatic copies of the 22 squares and 24 tetrahedra of

Lemma 38. With $X_j(s) = \sum_{\{p_u,p_v\}\in\mathcal{O}_j} s(p_u)s(p_v)$, set

$$\widetilde{\mathcal{L}}_{\mathcal{S},\mathcal{T}}(s) = c + \sum_{j=0}^{55} b_j X_j(s),$$

$$\widetilde{K}_{\mathcal{S},\mathcal{T}}(t) = c + \sum_{j=0}^{55} |\mathcal{O}_j| b_j \Omega_3(\sqrt{q_j}\, t).$$

The LP (with solver bounds $|c|, |b_j| \le 50$ and sample grid $\mathcal{N}_{\mathcal{S},\mathcal{T}}$) is

$$\begin{array}{ll} \text{maximize} & \delta \\ \text{variables} & \delta, c, (b_j)_{0\le j\le 55} \\ \text{subject to} & \widetilde{\mathcal{L}}_{\mathcal{S},\mathcal{T}}(s) \ge 0 \quad \text{for } s \in \mathcal{A}_{\mathcal{S},\mathcal{T}}, \\ & \widetilde{K}_{\mathcal{S},\mathcal{T}}(t) \le -\delta \quad \text{for } t \in \mathcal{N}_{\mathcal{S},\mathcal{T}}. \end{array}$$

Its rounded solution $c = -36608/1000$ and $b_j = a_j/1000$ gives $\mathcal{L}_S(s)$ in (8.1). Table 8 summarizes the variable and constraint counts for all eight LPs.

10. Finite subconfigurations and open problems

In each of Theorems 1–9, the space $\mathcal{X} \subseteq \{1,\dots,k\}^{\mathbb{F}^d}$ of valid colorings of $\mathbb{F}^d$ is empty, so compactness of $\{1,\dots,k\}^{\mathbb{F}^d}$ reduces the theorem to a finite subset $V \subset \mathbb{F}^d$:

Corollary 42. *For each of Theorems 1–9, the Ramsey property holds on a finite subset $V \subset \mathbb{F}^d$, and hence on every closed ball $B(0,R) \subset \mathbb{E}^d$ of sufficiently large radius R.*

Proof. Let $\mathcal{C}_\alpha$ be the closed cylinder of colorings that avoid a specified monochromatic occurrence on the finite vertex set $V_\alpha \subset \mathbb{F}^d$. Then the space of admissible colorings is $\mathcal{X} = \bigcap_{\alpha\in I} \mathcal{C}_\alpha = \emptyset$. Because $\{1,\dots,k\}^{\mathbb{F}^d}$ is compact, we can already find a finite subcollection $I_0 \subset I$ with

$$\bigcap_{\alpha\in I_0} \mathcal{C}_\alpha = \emptyset.$$

If we now collect all the underlying vertices into the finite set $V = \bigcup_{\alpha\in I_0} V_\alpha$, then no k-coloring of V can avoid our forbidden configurations, and it's clear that any Euclidean ball large enough to hold a translate of V inherits the same property. □

Note that we can freely rescale V by any $s > 0$ to get the same result at scale s, or embed V into $\mathbb{E}^n$ to handle any ambient dimension $n \ge d$.

Corollary 43. *In each setting of Theorems 1–9, any coloring of $\mathbb{E}^d$ must contain infinitely many pairwise disjoint monochromatic copies of a single configuration.*

Proof. We can simply pack $\mathbb{E}^d$ with an infinite family of disjoint balls of radius R, where R is chosen via Corollary 42. Each ball contains one of the finitely many forbidden monochromatic configurations. By the pigeonhole principle, a single color–configuration pair occurs in infinitely many of these balls, and the copies are disjoint. □

Because analogous LP searches did not give valid inequalities for (ℓ_2, ℓ_8) in $\mathbb{E}^2$, for the line problems in lower dimensions, or for (ℓ_2, P_7) and $(\mathcal{S},\mathcal{S})$ in $\mathbb{E}^3$, we ask:

Problem 1. *Does $\mathbb{E}^2 \to (\ell_2, \ell_8)$ hold?*

Problem 2. *Can any of the dimensions in Theorems 3–6 be lowered?*

Problem 3. *Does* $\mathbb{E}^3 \to (\mathcal{S}, \mathcal{S})$ *hold?*

Problem 4. *Does* $\mathbb{E}^2 \to (\ell_2, P_7)$ *hold?*

Problem 5. *For each relation in Corollary 42, what is the infimum of the radii of balls on which it holds?*

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Appendix A. Computational verifications

The code used for the computational verifications is available at `https://github.com/pretzeledkoala/measurable-lines`.

Numerical estimates use intervals rounded outwards. Exhaustive coloring checks use the listed integers or exact terminating decimals.

A.1. **Linear-program sampling data.** The kernel sample sets used in the linear programs are

$$\begin{aligned} \mathcal{N}_7 &= \{j/400 : 0 \le j \le 40000\}, & \mathcal{N}_{\mathrm{col}} &= \{j/400 : 0 \le j \le 10000\}, \\ \mathcal{N}_1 &= \{j/5000 : 0 \le j \le 30000\}, & \mathcal{N}_2 &= \{j/2000 : 0 \le j \le 50000\}, \\ \mathcal{N}_3 &= \{j/10000 : 0 \le j \le 100000\}, & \mathcal{N}_4 &= \{j/2500 : 0 \le j \le 62500\}, \\ \mathcal{N}_{\mathcal{S},\mathcal{T}} &= \{j/1000 : 0 \le j \le 40000\} \end{aligned}$$

where $j \in \mathbb{Z}$. For (ℓ_2, P_5), the LP uses 2140 pairs (p, t) to select the coefficients, and Appendix A.5 checks the resulting kernel K_{P_5}; for W_4, Appendix A.3 checks $[0, 125]$ on 2500 intervals of width $1/20$; and for $\mathcal{S}$–$\mathcal{T}$, the LP includes box bounds $|c|, |b_j| \le 50$.

Using $J_0(x) = \pi^{-1} \int_0^\pi \cos(x \sin\theta)\, d\theta$ [28, Eq. 10.9.1] and Hankel's expansion with remainder [28, Eq. 10.17.3 and Sec. 10.17(iii)], we record the analytic bounds used below.

Lemma 44.

(i) $|J_0'(x)| < 2/3$ *for* $x \in \mathbb{R}$.
(ii) $|J_0(x)| \le \sqrt{2/(\pi x)}$ *for* $x \ge 2$.
(iii) $|J_0(\sqrt{q}t)| < 1/(6q^{1/4})$ *for* $q \ge 1$ *and* $t \ge 25$.
(iv) $|J_0(\sqrt{q}t)| < 5/(6\sqrt{t}\, q^{1/4})$ *for* $q \ge 1$ *and* $t \ge 100$.

(v) $|\Omega_3(t)| \le 1/t$ *for* $t > 0$.
(vi) $|\Omega_4(t)| \le 2/t$ *for* $t > 0$.
(vii) $|\Omega_7(t)| \le 30/t^3$ *for* $t \ge 10$.

Proof. From $J_0(x) = \pi^{-1}\int_0^\pi \cos(x\sin\theta)\,d\theta$, differentiation under the integral sign gives

$$|J_0'(x)| \le \frac{1}{\pi}\int_0^\pi \sin\theta\,d\theta = \frac{2}{\pi} < \frac{2}{3}.$$

For (ii), set $\chi = x - \pi/4$ and write

$$J_0(x) = \sqrt{\frac{2}{\pi x}}\Big(A(x)\cos\chi + B(x)\sin\chi\Big),$$

where $A = 1 - 9/(128x^2) + R_e$ and $B = 1/(8x) - R_o$ with $0 \le R_e \le 0.113x^{-4}$ and $0 \le R_o \le 0.074x^{-3}$. When $x \ge 2$, we have $0 < 1 - 9/(128x^2) \le A \le 1 - 0.04x^{-2}$ (as $0.113/4 < 9/128 - 0.04$) and $0 \le B \le 1/(8x)$ (as $0.074 < 0.125x^2$), so

$$A^2 + B^2 \le (1 - 0.04x^{-2})^2 + 0.015625x^{-2} = 1 - 0.064375x^{-2} + 0.0016x^{-4} \le 1,$$

and Cauchy–Schwarz gives (ii). Putting $x = \sqrt{q}\,t$ in (ii) with $\pi > 3$ gives (iii)–(iv); $\Omega_3(t) = (\sin t)/t$ gives (v); $|J_1(t)| \le 1$ [28, Eq. 10.9.2] gives (vi); and $|\Omega_7(t)| \le 15(t^2+3t+3)/t^5 \le 30/t^3$ $(t \ge 10)$ gives (vii). □

For $z \in \mathbb{R}$ and $u = z^2/4$, truncating $J_0(z) = \sum_{m=0}^\infty (-1)^m u^m/(m!)^2$ at $m = N$ when $u < (N+2)^2$ gives

$$\left|\sum_{m=N+1}^{\infty} \frac{(-1)^m u^m}{(m!)^2}\right| \le \frac{u^{N+1}}{((N+1)!)^2}\,\frac{1}{1 - u/(N+2)^2}. \tag{A.1}$$

When $q, t \in \mathbb{Q}_{\ge 0}$ and $z = \sqrt{q}\,t$, one can show that the parameter $u = qt^2/4$, the Nth partial sum, and (A.1) are all rational. Let I be an interval on which $|K'| \le D$. Suppose that for every $t \in I$ there is a sample point $s \in I$ with $|t - s| \le h/2$ and $K(s) \le M$. Then the mean value theorem gives $K(t) \le M + Dh/2$ for every $t \in I$.

A.2. **Local coloring enumerations.** On the 18-point strip P, setting $r_p(R) = \mathbf{1}_{p\in R}$ gives $F(R) = -1 + \sum_{p\in P} c_p r_p(R) + \sum_{\{p,q\}\subset P} a_{pq} r_p(R) r_q(R) \ge 0$ on all 751 subsets $R \subseteq P$ avoiding the 33 unit edges and meeting all six lines $P_{i,s}$. On the 9-point grid Λ ($g = \mathbf{1}_G$, $s = \mathbf{1}_R - \mathbf{1}_B$), we have $4\mathcal{L} \ge 0$ on all 304 valid three-colorings; for W_4 and P_5, we have $10^6\mathcal{L}_{W_4} \ge 0$ on all 2338 valid colorings and $\mathcal{L}_{P_5} \ge 0$ on all 42 valid colorings. For the four line graphs $(\mathcal{V}_q, \mathcal{E}_q)$ $(m_q = 3, 6, 6, 4)$:

case	excluded consecutive patterns	vertices	edges
(ℓ_3, ℓ_4)	111, 0000	7	12
(ℓ_3, ℓ_5)	111, 00000	41	73
(ℓ_3, ℓ_6)	111, 000000	43	78
(ℓ_4, ℓ_4)	1111, 0000	14	26

TABLE 3. Vertex and edge counts for the four line graphs.

and the potentials ϕ_q of Appendix B satisfy $w_q(u,v) + \phi_q(u) - \phi_q(v) \ge 0$ on all $|\mathcal{E}_q|$ edges.

A.3. **Verification of the W_4 kernel bound.** Define $H(t) = K_{W_4}(t) - 600J_0(t) = \sum_q c_q J_0(\sqrt{q}\,t)$ using the coefficients from Table 11 (after subtracting 600 from c_1), and then partition $[0, 125]$ into 2500 intervals I of width $h = 1/20$ and midpoint m_I. We can now compute $u_I \geq H(m_I)$ and $d_I \geq |H'(m_I)|$, and truncate each J_0 or J_1 series only when the bound (A.1) is below 10^{-80}. Since $|J_0''| \leq 1$, we can apply Taylor's theorem to get

$$H(t) \leq u_I + \frac{h}{2} d_I + \frac{h^2}{8} \sum_q |c_q| q < 46 \qquad (t \in I).$$

This estimate holds on all 2500 intervals.

For the tail, we have $\sum_q |c_q| q^{-1/4} < 542$. By Lemma 44(ii) and $\pi > 3$,

$$|H(t)| \leq \sqrt{\frac{2}{3t}} \sum_q |c_q| q^{-1/4} < 40 \qquad (t \geq 125).$$

We now use the same coefficient tables to get $\sum_i \alpha_i = 139.366276$ and $\sum_{i<j} \beta_{ij} = 303.205864$, hence $A + 46 = 185.366276$ and $B - 646 = -342.794136$.

A.4. **Verification of the K_7 kernel bound.** Table 4 lists the pair count m_q and coefficient sum A_q for each squared distance in

$$\mathcal{Q} = \{1, 3, 4, 7, 9, 13, 16, 21, 25, 31, 36, 43, 49, 57\},$$

where $K_7(t) = -319 + \sum_{q \in \mathcal{Q}} A_q J_0(\sqrt{q}\,t)$; the classes $q = 64, 73$ have $A_q = 0$. Using

q	m_q	A_q
1	33	−489
3	15	1035
4	14	−669
7	13	600
9	12	−700
13	11	548
16	10	−355
21	9	441
25	8	−105
31	7	339
36	6	−114
43	5	−139
49	4	16
57	3	−90

TABLE 4. Distance classes and coefficient sums for K_7.

$|K_7'| \leq D := 11682.216002$ (from Lemma 44(i)), we step across $0 = t_0 < t_1 < \cdots < t_L = 100$: given an upper bound $K_i^+ \geq K_7(t_i)$ with $K_i^+ < 0$, choosing t_{i+1} with $D(t_{i+1} - t_i) \leq -K_i^+/2$ ensures $K_7(t) \leq K_i^+ + D(t - t_i) \leq K_i^+/2 < 0$ on $[t_i, t_{i+1}]$, which reaches $t_L = 100$. For $t \geq 100$, $\sum_{q \in \mathcal{Q}} |A_q| q^{-1/4} < 3600$ and Lemma 44(iv) give $K_7(t) < -319 + 3000/\sqrt{t} \leq -19$.

A.5. **Verification of the three-color and pentagon kernels.** With grid spacing h, sample maximum M, and derivative bound $|K'| \le D$ (from Lemma 44(i)), the bound $M + Dh/2$ on bounded intervals is summarized in Table 5:

kernel	interval	h	M	D	interval bound
H_g	$[0,25]$	0.0025	180	10151	192.69
H_s	$[0,25]$	0.0025	184.076	2208	186.84
K_{P_5}	$[0,12]$	0.005	389	1330	392.33

TABLE 5. Sample, derivative, and interpolation bounds for H_g, H_s, and K_{P_5}.

For H_g and H_s (sampled at $t_j = j/400 \in [0,25]$), Table 5 gives $H_g, H_s < 193$ on $[0,25]$; for $t \ge 25$, Lemma 44(iii) with $3^{-1/4} < 0.77$, $4^{-1/4} < 0.715$, $7^{-1/4} < 0.625$, $12^{-1/4} < 0.556$ gives $H_g(t) < 55$ and $H_s(t) < 143$. For K_{P_5} (sampled at the 2400 cell midpoints $t_j = (2j+1)/400 \in [0,12]$), Table 5 gives $K_{P_5}(t) < 393$ on $[0,12]$; for $t \ge 12$, writing $(q_1, q_2, q_3, q_4) = ((3-\sqrt{5})/2, 1, (5-\sqrt{5})/2, (3+\sqrt{5})/2)$ and $(c_1, c_2, c_3, c_4) = (200, -1048, 100, -385)$, the bounds $q_1^{1/4} > 0.78$, $q_3^{1/4} > 1.08$, $q_4^{1/4} > 1.27$ give $\sum_{i=1}^4 |c_i| q_i^{-1/4} < 1701$, whence $|K_{P_5}(t)| < 8505/\sqrt{471} < 392$ by Lemma 44(ii) (as $\sqrt{q_1}t > 2$ and $\pi > 3.14$).

A.6. **Verification of the four line kernels.** From Lemma 15, we have $|\Omega_3'| \le 1/2$, $|\Omega_4'| \le 4/(3\pi)$, and $|\Omega_7'| \le 5/16$, which give the derivative bounds D and mean-value bounds $M + Dh/2$ in Table 6 (where each grid of step h includes both endpoints): On

kernel	interval	h	M	D	interval bound
H_1	$[0,6]$	0.0002	0.69	3.5	0.6904
H_2	$[0,25]$	0.0005	2.938	47.5	2.9499
H_3	$[0,10]$	0.0001	0.399	11.875	0.3996
H_4	$[0,25]$	0.0004	0.887	12	0.8894

TABLE 6. Sample, derivative, and interpolation bounds for the four line kernels.

the complementary rays, Lemma 44(v)–(vii) bounds $|c_k\Omega_3(kt)| \le |c_k|/(kt)$, $|c_k\Omega_4(kt)| \le 2|c_k|/(kt)$, and $|c_k\Omega_7(kt)| \le 30|c_k|/(kt)^3$ ($t \ge 10$), giving Table 7 and completing the proof of Proposition 27.

kernel	range	tail bound	bound on the range
H_1	$t \ge 6$	$4/t$	0.667
H_2	$t \ge 25$	$60/t$	2.4
H_3	$t \ge 10$	$291/t^3$	0.291
H_4	$t \ge 25$	$18/t$	0.72

TABLE 7. Tail bounds for the four line kernels.

A.7. **Computational verification for the square–tetrahedron configuration.** The geometric and local counts in Section 8 follow from arithmetic in $\mathbb{Q}(\sqrt{3})$. Represent $a+b\sqrt{3}$ by (a,b), with addition and multiplication given by $(a,b)+(c,d)=(a+c,b+d)$ and $(a,b)(c,d)=(ac+3bd,ad+bc)$. We now compute $d_{ij}=|p_i-p_j|^2$ between the 20 points $p_0,\dots,p_{19}$ (by brute force). Among the $\binom{20}{4}=4845$ four-point subsets, 46 have squared distances $\{1,1,1,1,2,2\}$. Of these, 22 are copies of $\mathcal{S}$ and 24 are copies of $\mathcal{T}$, depending on whether the two longest edges are disjoint or share a vertex.

The group $G=\langle\tau,\sigma\rangle$ has four elements. Its action on the unordered pairs is obtained from

$$\mathcal{O}(i,j)=\Big\{\{\min(g(i),g(j)),\max(g(i),g(j))\}:g\in G\Big\},$$

and a calculation gives the following orbit-size distribution:

orbit size	1	2	4	total
number of orbits	2	14	40	56
pairs accounted for	2	28	160	190

The representatives and the associated distances and coefficients are listed in Appendix B.

Substituting the integer coefficients from Appendix B into the local expression gives $1000\mathcal{L}_S\geq 0$ for every allowed coloring.

On the grid $t_j=j/1000$ in $[0,40]$, interval evaluation gives $K_S(t_j)<-0.988$. Every squared distance in the kernel is less than 9, and Lemma 15 gives $|\Omega_3'|\leq 1/2$, so $|K_S'(t)|<931$. Each point of $[0,40]$ is within $1/2000$ of a grid point; hence $K_S(t)<-0.988+0.4655=-0.5225<-1/2$.

For $t\geq 40$, the smallest squared distance is $2-\sqrt{3}>1/4$, so $|\Omega_3(\sqrt{q}\,t)|\leq 2/t$. Therefore $K_S(t)\leq -36.608+1240.536/t\leq -5.5946<-1/2$.

A.8. **Linear-program sizes.** Table 8 records the dimensions and scalar constraint counts of the eight linear programs used in the coefficient searches. The W_4 local and kernel verifications are recorded separately in this appendix. The columns labeled local, kernel, and variable bounds count scalar inequalities. The equality column counts scalar equalities. The total number of constraints is the sum of these four columns. A two-sided variable bound gives two scalar inequalities.

LP	variables	local	kernel	equalities	variable bounds	total constraints
(ℓ_2,ℓ_7)	$172=1+18+153$	751	40001	0	342	41094
(ℓ_2,ℓ_2,ℓ_3)	$31=1+4+13+13$	304	20002	0	0	20306
(ℓ_3,ℓ_4)	$11=1+1+2+7$	12	30001	1	22	30036
(ℓ_3,ℓ_5)	$49=1+1+6+41$	73	50001	1	98	50173
(ℓ_3,ℓ_6)	$50=1+1+5+43$	78	100001	1	100	100180
(ℓ_4,ℓ_4)	$19=1+4+14$	26	62501	1	38	62566
(ℓ_2,P_5)	$47=2+9+36$	42	2140	0	94	2276
$\mathcal{S}$–$\mathcal{T}$	$58=1+1+56$	47946	40001	0	114	88061

TABLE 8. Variables and constraints in the linear programs.

Appendix B. Coefficient data

This appendix records the coefficient values used in the local inequalities. Together with the definitions and verifications in Appendix A, these data suffice to check the local inequalities and their averaged bounds.

The ordering conventions are fixed as follows. In the nine-point configuration, order point labels lexicographically and take the lexicographically least label in each vertex orbit, or the lexicographically least ordered pair in each unordered pair orbit, as its representative. The table rows follow that order. In the eighteen-point configuration, use the point order $00, 01, \dots, 08, 10, 11, \dots, 18$. The rows and columns of each coefficient array are indexed by $0, 1, \dots, 8$, and the squared-distance table is in increasing order. For the square–tetrahedron configuration, the orbit $\mathcal{O}_j$ is the orbit represented in row j of the table below, and that row order defines the indices used in X_j, a_j, and B_q.

B.1. **The W_4 inequality.** For the point order fixed in Section 6, the pairs forming ℓ_2 are

$$\begin{aligned}\mathcal{E}_4 = \{&(0,5),(0,15),(1,2),(1,4),(1,6),(1,16),\\ &(2,3),(2,7),(2,17),(3,4),(3,8),(3,18),\\ &(4,9),(4,19),(5,10),(6,7),(6,9),(6,11),\\ &(7,8),(7,12),(8,9),(8,13),(9,14),(10,15),\\ &(11,12),(11,14),(11,16),(12,13),(12,17),(13,14),\\ &(13,18),(14,19),(16,17),(16,19),(17,18),(18,19)\}.\end{aligned}$$

The notation (i,j) denotes the unordered pair $\{i,j\}$.

The dihedral symmetries of the unit square act on the twenty-point configuration in the main text. To avoid repeated denominators, write

$$10^6\mathcal{L}_{W_4}(r) = C + \sum_i A_i r_i + \sum_{i<j} B_{ij} r_i r_j.$$

The constant is $C = -24994229$, and the tables below give the integers A_i and B_{ij}. The linear coefficients are constant on the following three vertex orbits.

indices	orbit size	A_i
$0, 5, 10, 15$	4	3167237
$1, 2, 7, 8, 13, 14, 16, 19$	8	5837166
$3, 4, 6, 9, 11, 12, 17, 18$	8	10000000

Table 9. Linear coefficients in the W_4 inequality.

For pairs, the first column gives the representative of a pair orbit under these symmetries, the second gives its size, and the last gives the common scaled integer B_{ij}. The corresponding coefficient in $\mathcal{L}_{W_4}$ is $\beta_{ij} = 10^{-6}B_{ij}$. These rows determine the complete upper-triangular matrix (β_{ij}).

With the kernel bound $K_{W_4}(t) - 600J_0(t) < 46$, the probability bound is

$$\Phi_0(p) = -24.994229 + 185.366276p - 342.794136p^2.$$

representative	orbit size	B_{ij}
$(0,1)$	8	3129119
$(0,3)$	8	−10000000
$(0,5)$	4	10000000
$(0,6)$	8	−2904617
$(0,7)$	8	355813
$(0,8)$	8	−2904617
$(0,9)$	8	−618433
$(0,10)$	2	10000000
$(0,11)$	8	−487123
$(0,13)$	8	9381567
$(1,2)$	4	10000000
$(1,3)$	8	−7095383
$(1,4)$	8	10000000
$(1,6)$	8	10000000
$(1,7)$	8	0
$(1,8)$	4	−8479162
$(1,9)$	8	−487123
$(1,11)$	8	−487123
$(1,12)$	8	−487123
$(1,13)$	4	4294145
$(1,14)$	4	4162834
$(1,16)$	4	10000000
$(1,17)$	8	0
$(1,18)$	8	2286183
$(3,4)$	4	10000000
$(3,6)$	4	355813
$(3,9)$	8	487123
$(3,11)$	4	0
$(3,12)$	4	131310
$(3,18)$	4	10000000

TABLE 10. Pair coefficients in the W_4 inequality.

q	$\sum_{\substack{i<j \\ \lvert v_i - v_j \rvert^2 = q}} \beta_{ij}$
$\frac{1}{2}$	−54.967048
1	360
$\frac{3}{2} + \sqrt{2}$	−23.236936
$\frac{3}{2}$	−2.10096
$\frac{3}{2} - \sqrt{2}$	−23.236936
2	−40.660048
$\frac{5}{2} + \sqrt{2}$	−3.896984
$\frac{5}{2} - \sqrt{2}$	75.052536
$2 - \sqrt{2}$	−3.896984
$3 - 2\sqrt{2}$	−17.265312
3	14.39248
$4 - 2\sqrt{2}$	17.17658
$2 + \sqrt{2}$	3.896984
$3 + 2\sqrt{2}$	1.948492

TABLE 11. Nonzero coefficient sums for the W_4 kernel.

The proof rounds these coefficients upwards to obtain Φ. Direct evaluation gives $\max_{0 \le p \le 1/4} \Phi_0(p) < -0.07$.

B.2. **The nine-point inequality.** The group acting on the nine-point configuration has four vertex orbits and thirteen orbits of unordered pairs. Each pair orbit is determined by the representative in the table by applying the four symmetries of the configuration.

representative	orbit size	coefficient c_O
00	2	−252
01	4	−1004
02	2	998
11	1	−345

TABLE 12. Nine-point vertex coefficients.

representative	orbit size	squared distance	a_U	b_U
00, 01	4	1	−998	2
00, 02	4	4	−1000	0
00, 11	2	3	500	150
00, 12	4	7	1000	300
00, 22	1	12	375	200
01, 02	4	1	−446	−894
01, 10	2	1	2	502
01, 11	4	1	444	948
01, 12	2	3	662	−12
01, 20	4	3	448	0
01, 21	2	4	−264	−12
02, 11	2	1	−52	−1000
02, 20	1	4	112	0

TABLE 13. Nine-point pair coefficients.

B.3. **The eighteen-point inequality.** The pair multiplicities and coefficient sums are recorded in Table 4. The next table gives the vertex coefficients, followed by the three arrays containing all coefficients for pairs.

The vertex coefficients are

p	00	01	02	03	04	05	06	07	08
c_p	0	−54	−54	−16	−2	15	−46	−1	26
p	10	11	12	13	14	15	16	17	18
c_p	18	−16	−52	1	−18	−23	−44	−53	0

For pairs, write

$$a^{00}_{jk} = a_{(0,j),(0,k)},$$
$$a^{11}_{jk} = a_{(1,j),(1,k)},$$
$$a^{01}_{jk} = a_{(0,j),(1,k)}.$$

The two within-column upper-triangular arrays and the cross-column array are as follows. A dash denotes an unused diagonal or lower-triangular entry.

$j\backslash k$	0	1	2	3	4	5	6	7	8
0	–	−100	0	0	0	0	0	0	0
1	–	–	−100	−2	−11	−12	14	−16	11
2	–	–	–	−100	−54	−100	−58	−22	−43
3	–	–	–	–	−100	−95	−100	−60	−55
4	–	–	–	–	–	−100	−87	−92	−58
5	–	–	–	–	–	–	−100	−63	−58
6	–	–	–	–	–	–	–	−100	−42
7	–	–	–	–	–	–	–	–	100
8	–	–	–	–	–	–	–	–	–

TABLE 14. The coefficients a^{00}_{jk} for pairs in the first column.

$j\backslash k$	0	1	2	3	4	5	6	7	8
0	–	100	−38	−52	−52	−49	−43	5	0
1	–	–	100	−57	−77	−46	−13	−12	0
2	–	–	–	100	−77	−100	−57	20	0
3	–	–	–	–	100	−95	−100	−12	0
4	–	–	–	–	–	100	−60	−10	0
5	–	–	–	–	–	–	100	1	0
6	–	–	–	–	–	–	–	100	0
7	–	–	–	–	–	–	–	–	100
8	–	–	–	–	–	–	–	–	–

TABLE 15. The coefficients a^{11}_{jk} for pairs in the second column.

$j\backslash k$	0	1	2	3	4	5	6	7	8
0	−100	0	0	0	0	0	0	0	0
1	−100	−100	40	14	31	30	57	26	0
2	77	−100	−100	99	78	79	100	59	0
3	63	85	−100	−100	93	67	68	26	0
4	63	57	100	−100	−100	84	56	28	0
5	52	61	65	70	−100	−100	85	7	0
6	67	96	100	73	100	−89	100	39	0
7	−85	93	90	58	60	88	100	100	0
8	−90	−80	63	47	60	60	75	100	100

TABLE 16. The coefficients a_{jk}^{01} for pairs between the two columns.

B.4. **The pentagon inequality.** Use the vectors u_i, translation s, and point labels v_i from the pentagon construction in Section 7. Within the two pentagons, the side pairs are $(0,1),(1,2),(2,3),(3,4),(4,0)$ and $(1,5),(5,6),(6,7),(7,8),(8,1)$. For completeness, the calculation for pairs joining the two pentagons uses

$$R_5^2 = \frac{5+\sqrt{5}}{10},\ |u_i - (u_j + s)|^2 = |u_i - u_j - u_1 + u_0|^2,$$

and

$$\cos 0 = 1,\ \cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4},\ \cos\frac{4\pi}{5} = -\frac{\sqrt{5}+1}{4}.$$

For $i \in \{0,1,2,3,4\}$ and $j \in \{1,2,3,4\}$, the pairs at distance one joining the two pentagons are $(i,j) = (1,1),(1,4),(2,2),(3,3),(4,4)$. The first two are already the translated-pentagon sides $(1,5)$ and $(1,8)$. The other three give $(2,6),(3,7),(4,8)$. Thus the complete unit-pair list is

$$\mathcal{E}_5 = \{(0,1),(0,4),(1,2),(1,5),(1,8),(2,3),(2,6),\\ (3,4),(3,7),(4,8),(5,6),(6,7),(7,8)\}.$$

The point labels are ordered as in the pentagon configuration in the main text. The linear coefficient vector is

$$(\alpha_0,\ldots,\alpha_8) = (-23,77,-100,-23,-23,100,100,100,77).$$

The coefficients for pairs are given by the following upper-triangular matrix; blank entries are zero.

$i\backslash j$	0	1	2	3	4	5	6	7	8
0		−100	23	23	−100				23
1			−100	23	23	−100	−100	−100	−100
2				−100	23	77	−100	77	100
3					−100			−100	23
4									−100
5							−100	−100	−100
6								−48	−100
7									100
8									

TABLE 17. Coefficients for pairs in the pentagon inequality.

B.5. **The four line inequalities.** The sequence labels in the following tables are binary. For ϕ_1, ϕ_2, and ϕ_3 they are the values of the red indicator r_j. For ϕ_4, they are the values $\epsilon_j = r_j = (s_j + 1)/2$. When the local weight is evaluated, the same sequence is converted to signed values by $s_j = 2\epsilon_j - 1$. Thus the tables list only sequences satisfying the prescribed restrictions; these are the entire domains of the functions ϕ_q. The two short tables of state values are

sequence	000	001	010	011	100	101	110
ϕ_1	0	−1	−2	−2	−3	−3	−3

TABLE 18. Potential values for the (ℓ_3, ℓ_4) inequality.

seq.	ϕ_4	seq.	ϕ_4	seq.	ϕ_4	seq.	ϕ_4
0001	0	0010	−8	0011	−10	0100	−12
0101	−14	0110	−14	0111	−12	1000	−12
1001	−14	1010	−14	1011	−12	1100	−10
1101	−8	1110	0				

TABLE 19. Potential values for the (ℓ_4, ℓ_4) inequality.

The two longer tables of state values follow.

sequence	ϕ_2	sequence	ϕ_2
000010	−2	000011	0
000100	−8	000101	−8
000110	−6	001000	−14
001001	−14	001010	−14
001011	−14	001100	−13
001101	−12	010000	−20
010001	−20	010010	−20
010011	−20	010100	−22
010101	−20	010110	−20
011000	−24	011001	−22
011010	−20	011011	−18
100001	−26	100010	−26
100011	−26	100100	−30
100101	−28	100110	−28
101000	−28	101001	−28
101010	−26	101011	−26
101100	−29	101101	−27
110000	−30	110001	−30
110010	−28	110011	−28
110100	−26	110101	−26
110110	−24		

TABLE 20. Potential values for the (ℓ_3, ℓ_5) inequality.

sequence	ϕ_3	sequence	ϕ_3
000001	0	000010	−2
000011	−2	000100	−4
000101	−4	000110	−4
001000	−6	001001	−6
001010	−6	001011	−6
001100	−6	001101	−6
010000	−8	010001	−8
010010	−8	010011	−8
010100	−9	010101	−8
010110	−9	011000	−8
011001	−8	011010	−8
011011	−8	100000	−10
100001	−10	100010	−11
100011	−10	100100	−12
100101	−11	100110	−13
101000	−11	101001	−11
101010	−11	101011	−10
101100	−12	101101	−11
110000	−10	110001	−10
110010	−10	110011	−10
110100	−10	110101	−10
110110	−10		

TABLE 21. Potential values for the (ℓ_3, ℓ_6) inequality.

B.6. **The twenty-point square–tetrahedron inequality.** The point order used throughout the coefficient tables is as follows. Label the points by

$$
\begin{array}{llll}
p_0 = 0, & p_1 = w_0, & p_2 = v_0, & p_3 = v_0 + w_0, \\
p_4 = e, & p_5 = e + w_0, & p_6 = e + v_0, & p_7 = e + v_0 + w_0, \\
p_8 = w_1, & p_9 = v_1, & p_{10} = v_1 + w_1, & p_{11} = e + w_1, \\
p_{12} = e + v_1, & p_{13} = e + v_1 + w_1, & p_{14} = w_2, & p_{15} = v_2, \\
p_{16} = v_2 + w_2, & p_{17} = e + w_2, & p_{18} = e + v_2, & p_{19} = e + v_2 + w_2.
\end{array}
$$

On the labels, the generators act as

$$\sigma = (1\ 15)(2\ 14)(3\ 16)(5\ 18)(6\ 17)(7\ 19)(8\ 9)(11\ 12),$$
$$\tau = (0\ 4)(1\ 5)(2\ 6)(3\ 7)(8\ 11)(9\ 12)(10\ 13)(14\ 17)(15\ 18)(16\ 19).$$

In Table 23, the representative is a pair of point labels from the square–tetrahedron section. The third column is the size of its orbit under the group G, the fourth is the common squared distance, and the last is the coefficient a_j in the square–tetrahedron expression.

q	B_q	q	B_q
$2-\sqrt{3}$	-68300	1	-300708
$3-\sqrt{3}$	56588	2	-142872
$4-\sqrt{3}$	-14632	3	18334
$5-\sqrt{3}$	10996	$2+\sqrt{3}$	628
4	-5700	$3+\sqrt{3}$	-404
$4+\sqrt{3}$	204	6	386
$5+\sqrt{3}$	380	7	136

TABLE 22. Coefficient sums at each squared distance in the square–tetrahedron kernel.

TABLE 23. Pair orbits and integer coefficients for the square–tetrahedron inequality.

j	representative	$\lvert\mathcal{O}_j\rvert$	q_j	a_j
0	$\{8,10\}$	4	1	0
1	$\{0,8\}$	4	1	-17344
2	$\{0,10\}$	2	2	-152
3	$\{0,1\}$	4	1	-9839
4	$\{4,8\}$	4	2	-21087
5	$\{0,2\}$	4	1	-10680
6	$\{4,10\}$	2	3	587
7	$\{0,3\}$	4	2	-2319
8	$\{3,4\}$	4	3	2040
9	$\{1,4\}$	4	2	-553
10	$\{1,3\}$	4	1	3317
11	$\{2,3\}$	4	1	-997
12	$\{7,9\}$	4	$3-\sqrt{3}$	5100
13	$\{0,4\}$	1	1	-39099
14	$\{5,14\}$	4	4	-1425
15	$\{2,4\}$	4	2	-1277
16	$\{2,14\}$	2	$2+\sqrt{3}$	-12
17	$\{1,14\}$	4	3	1304
18	$\{3,14\}$	4	$4+\sqrt{3}$	51
19	$\{6,14\}$	2	$3+\sqrt{3}$	-22
20	$\{1,9\}$	4	$2-\sqrt{3}$	-6903
21	$\{7,14\}$	4	$5+\sqrt{3}$	95
22	$\{1,2\}$	4	2	-1171
23	$\{5,9\}$	4	$3-\sqrt{3}$	5627
24	$\{1,5\}$	2	1	-6775
25	$\{5,10\}$	4	$3-\sqrt{3}$	485
26	$\{2,5\}$	4	3	1144
27	$\{2,9\}$	4	1	-9068
28	$\{1,8\}$	4	1	-2603
29	$\{3,5\}$	4	2	-2761
30	$\{3,15\}$	4	$4-\sqrt{3}$	-3658
31	$\{1,10\}$	4	$2-\sqrt{3}$	-337
32	$\{3,16\}$	2	6	193
33	$\{7,15\}$	4	$5-\sqrt{3}$	2749
34	$\{7,16\}$	2	7	68
35	$\{2,10\}$	4	$2+\sqrt{3}$	-294
36	$\{3,9\}$	4	$2-\sqrt{3}$	-6010
37	$\{3,7\}$	2	1	-2480
38	$\{9,11\}$	2	3	460
39	$\{5,8\}$	4	2	-5565
40	$\{1,15\}$	2	$2-\sqrt{3}$	-7650
41	$\{6,8\}$	4	$3+\sqrt{3}$	-21
42	$\{7,8\}$	4	$3+\sqrt{3}$	-145
43	$\{10,11\}$	4	2	537
44	$\{2,8\}$	4	$2+\sqrt{3}$	-93
45	$\{3,10\}$	4	2	-59
46	$\{3,6\}$	4	2	1433
47	$\{6,9\}$	4	2	-2065
48	$\{3,8\}$	4	$2+\sqrt{3}$	550
49	$\{5,15\}$	2	$3-\sqrt{3}$	5870
50	$\{6,10\}$	4	$3+\sqrt{3}$	76
51	$\{8,9\}$	2	2	-1510
52	$\{7,10\}$	4	3	-428
53	$\{10,13\}$	1	1	-267
54	$\{2,6\}$	2	1	-54
55	$\{8,11\}$	2	1	-26934

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